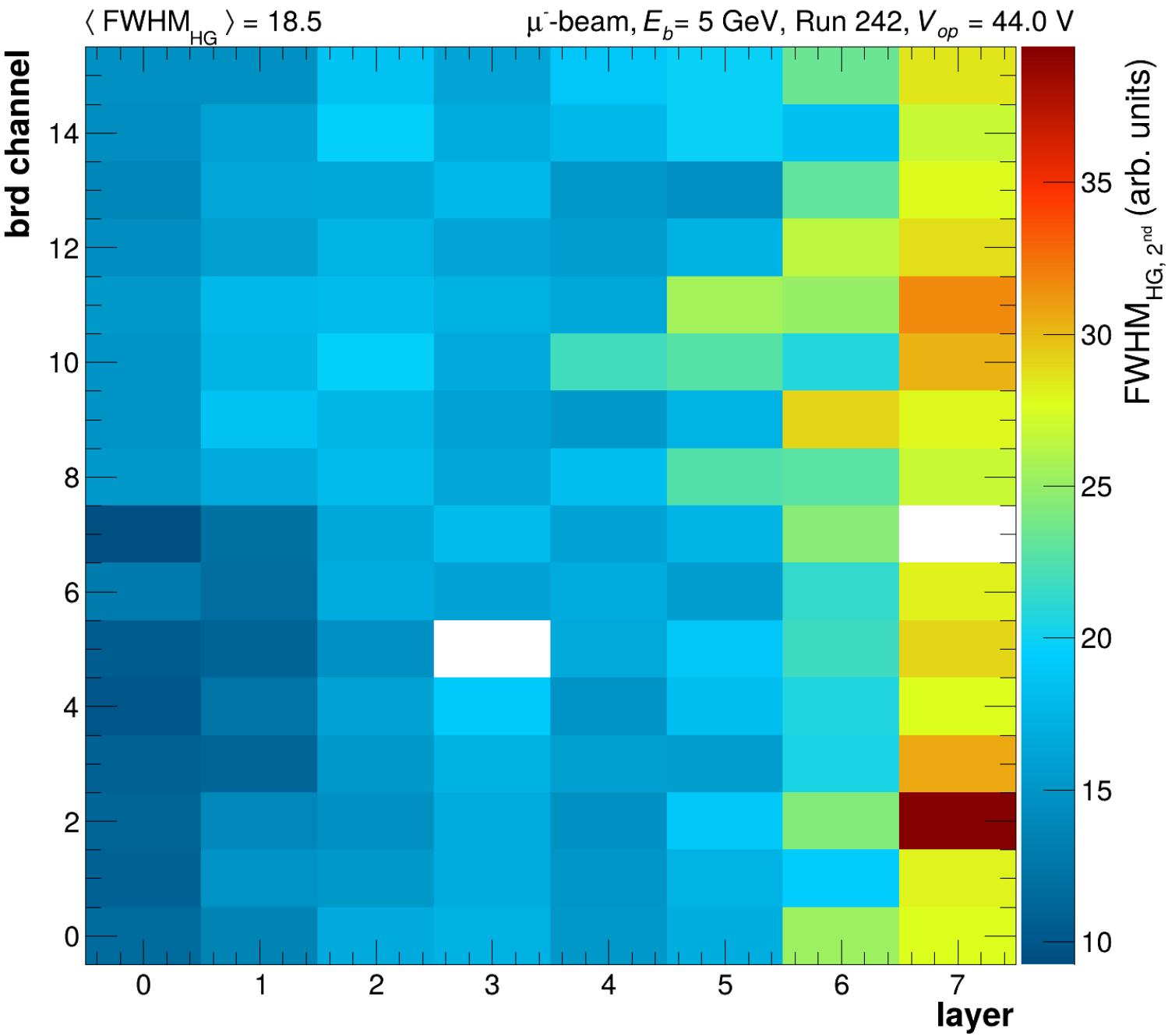
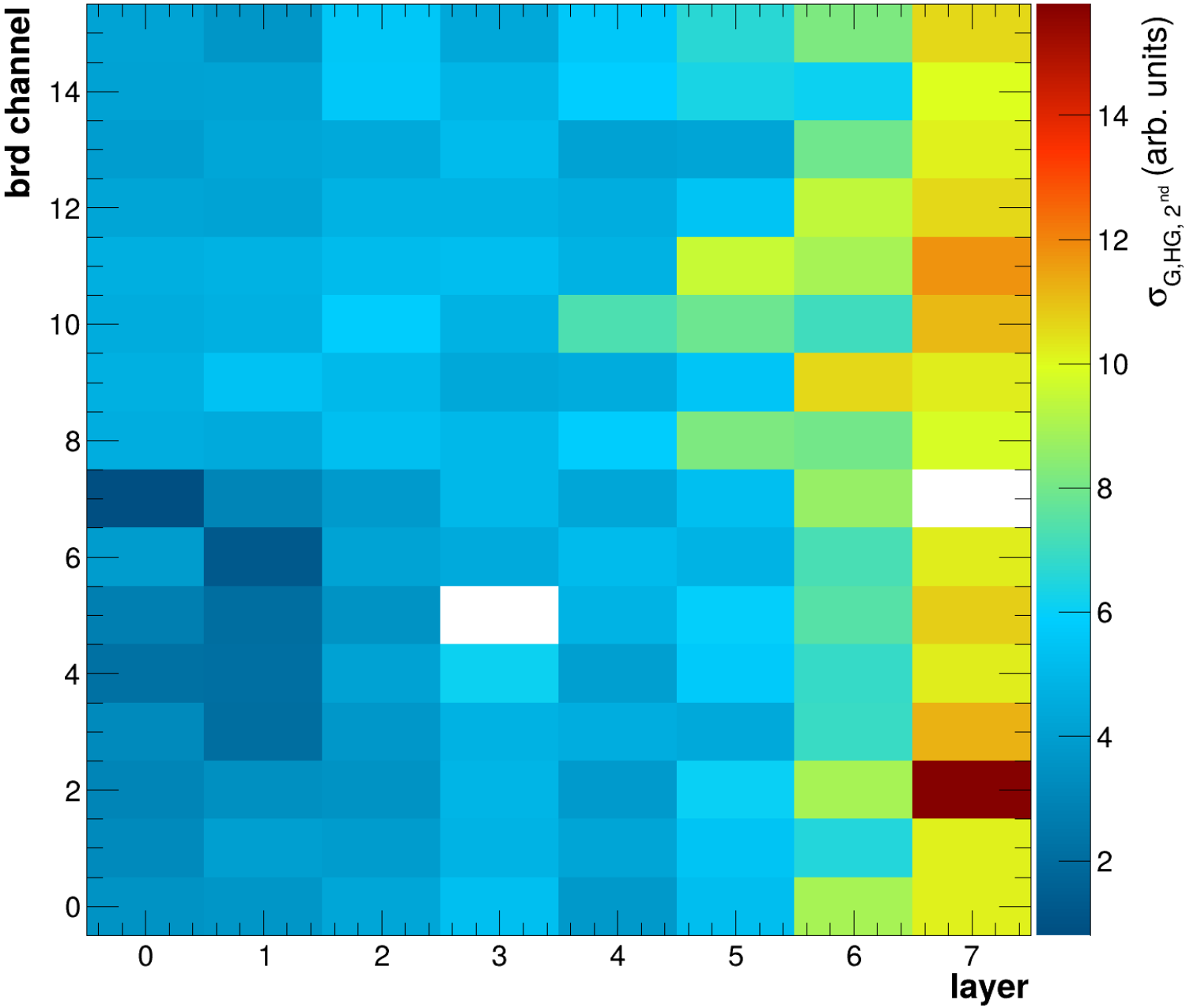
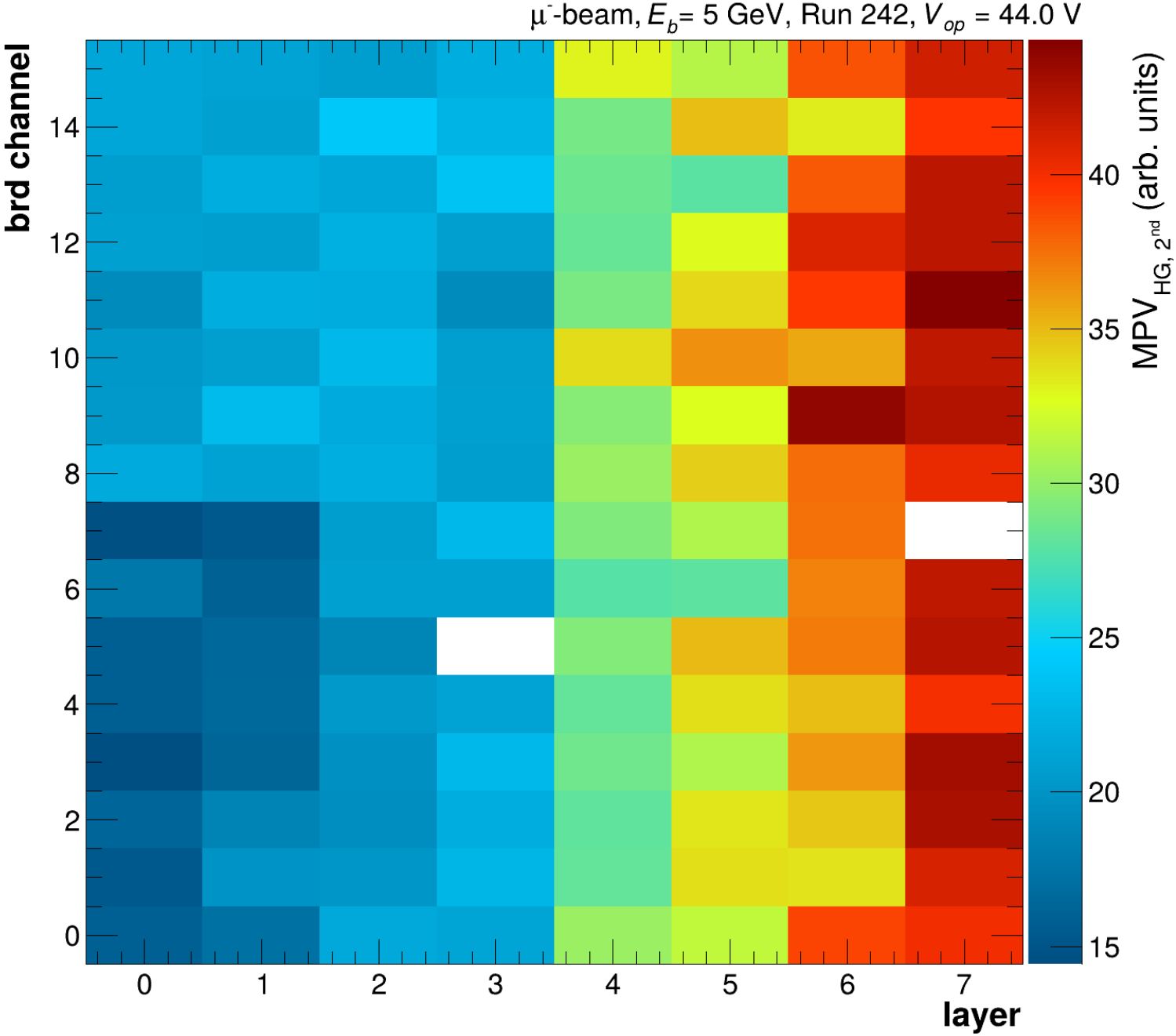
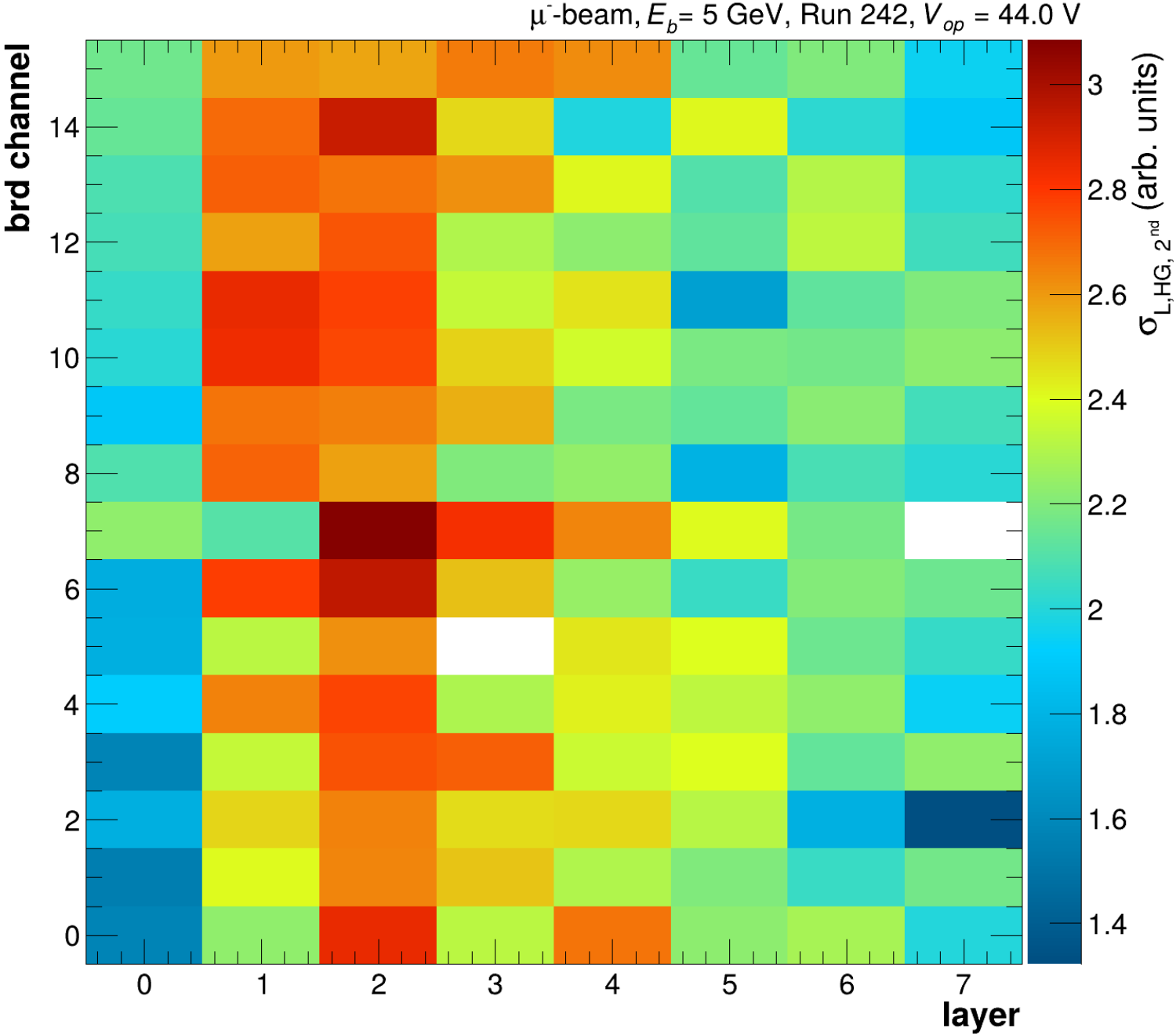
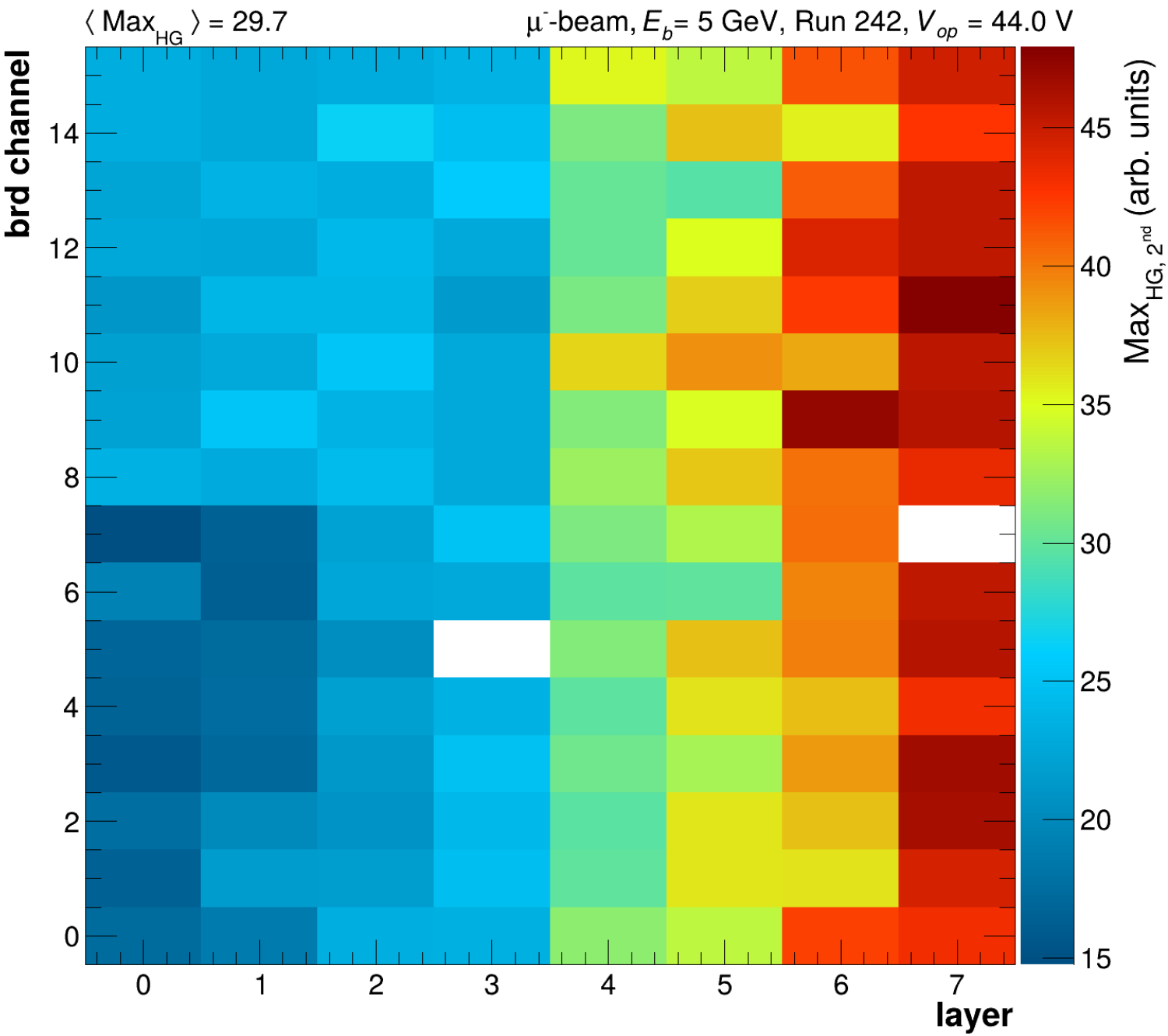
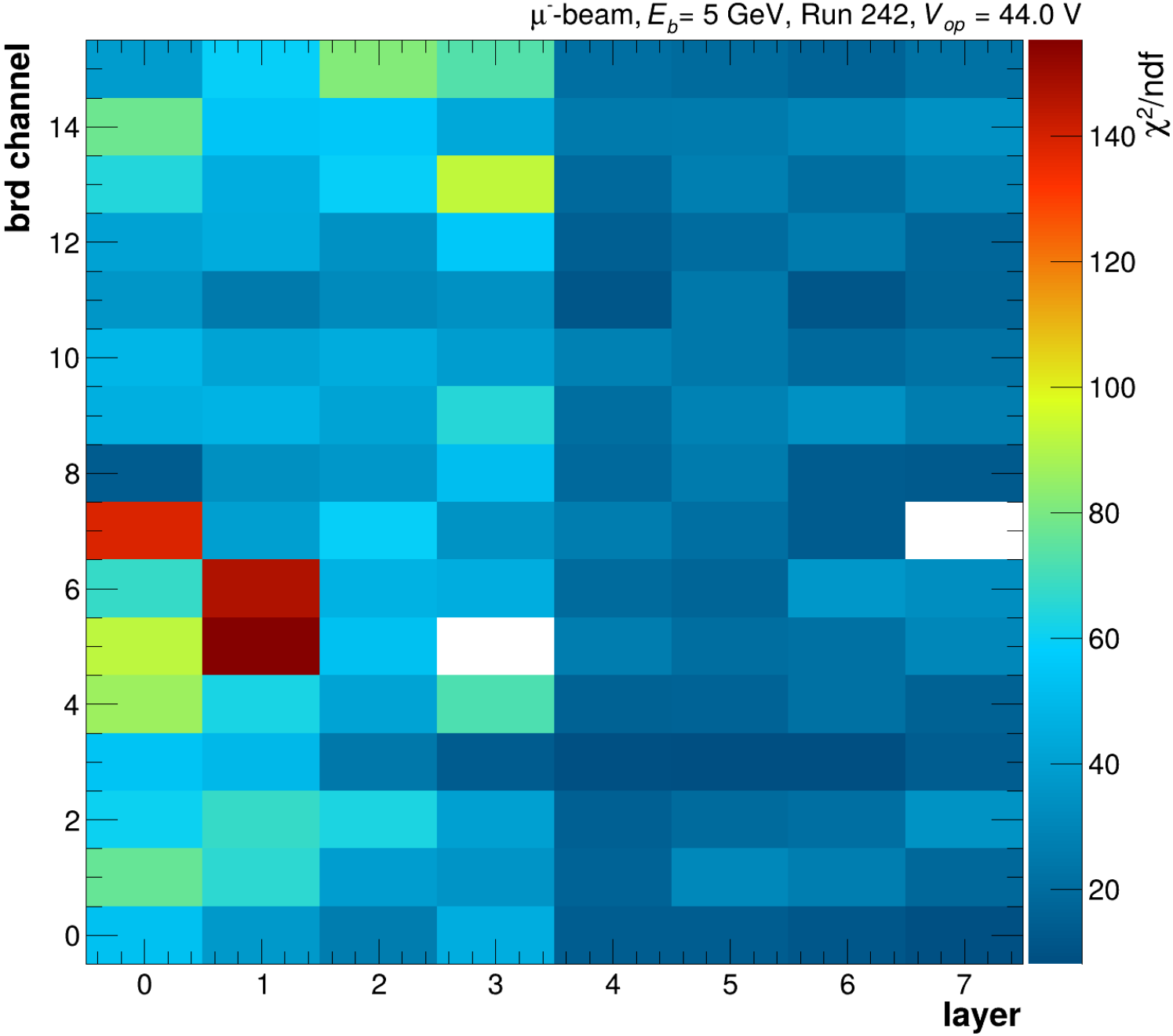

⟨ FWHM_HG ⟩ = 18.5
μ⁻-beam, E_b = 5 GeV, Run 242, V_op = 44.0 V
brd channel
layer
FWHM_HG, 2nd (arb. units)
0
2
4
6
8
10
12
14
0
1
2
3
4
5
6
7
10
15
20
25
30
35

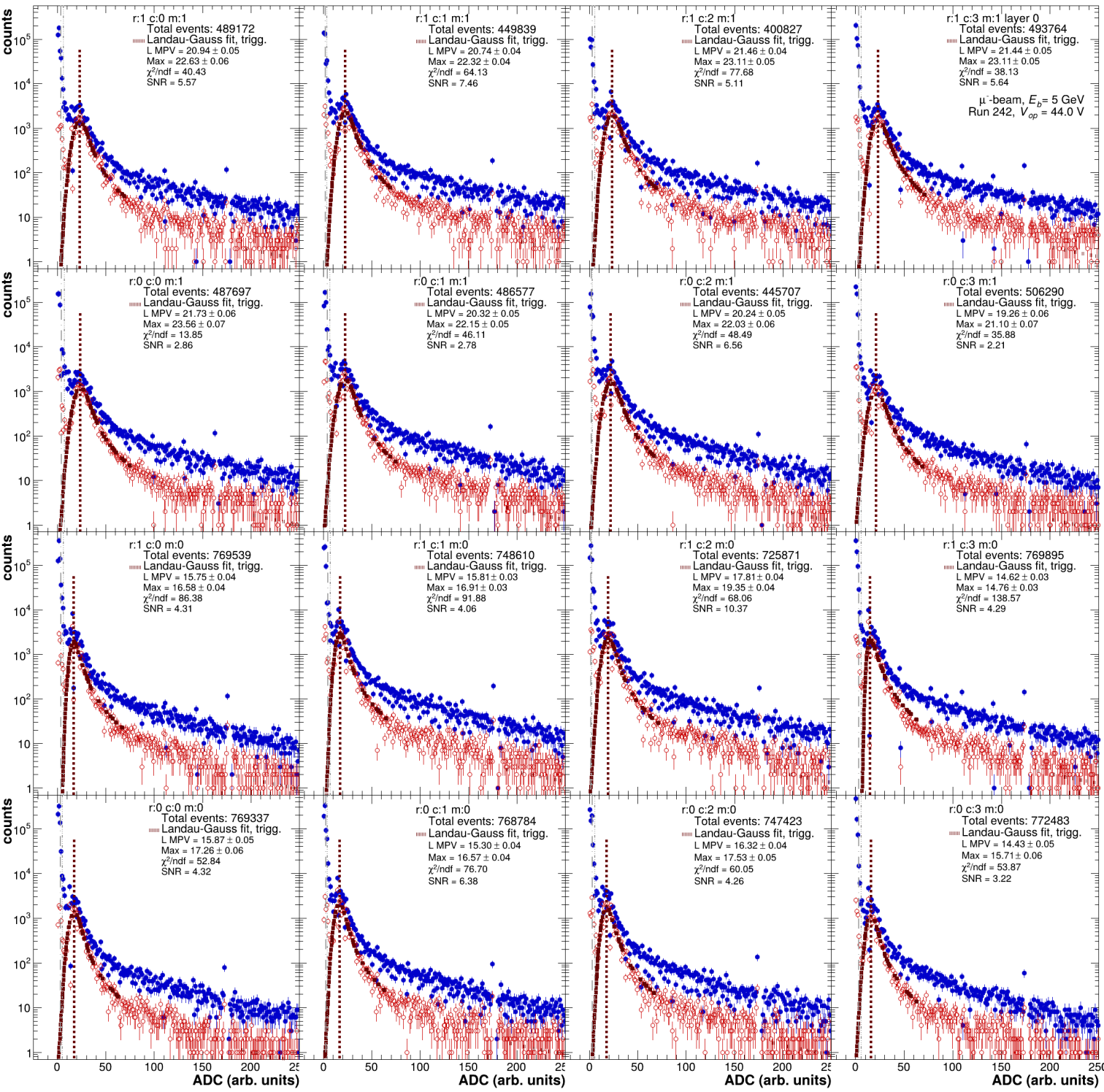

μ⁻-beam, E_b= 5 GeV, Run 242, V_{op} = 44.0 V
brd channel
layer
$\sigma_{G,HG,2^{nd}}$ (arb. units)
0
1
2
3
4
5
6
7
0
2
4
6
8
10
12
14
2
4
6
8
10
12
14

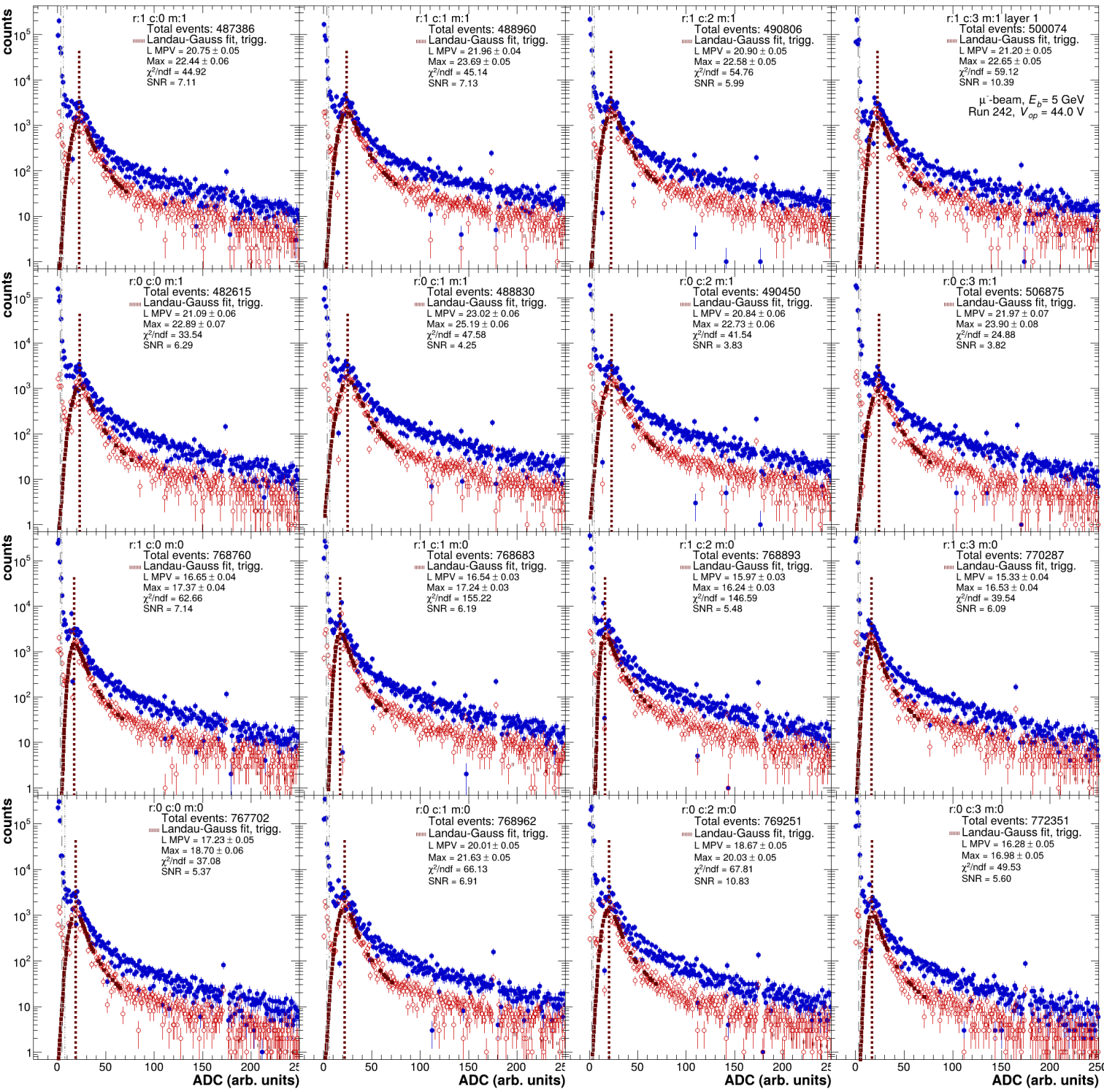

μ⁻-beam, E_b= 5 GeV, Run 242, V_{op} = 44.0 V
brd channel
0
2
4
6
8
10
12
14
0
1
2
3
4
5
6
7
layer
$MPV_{HG,\,2^{nd}}$ (arb. units)
15
20
25
30
35
40

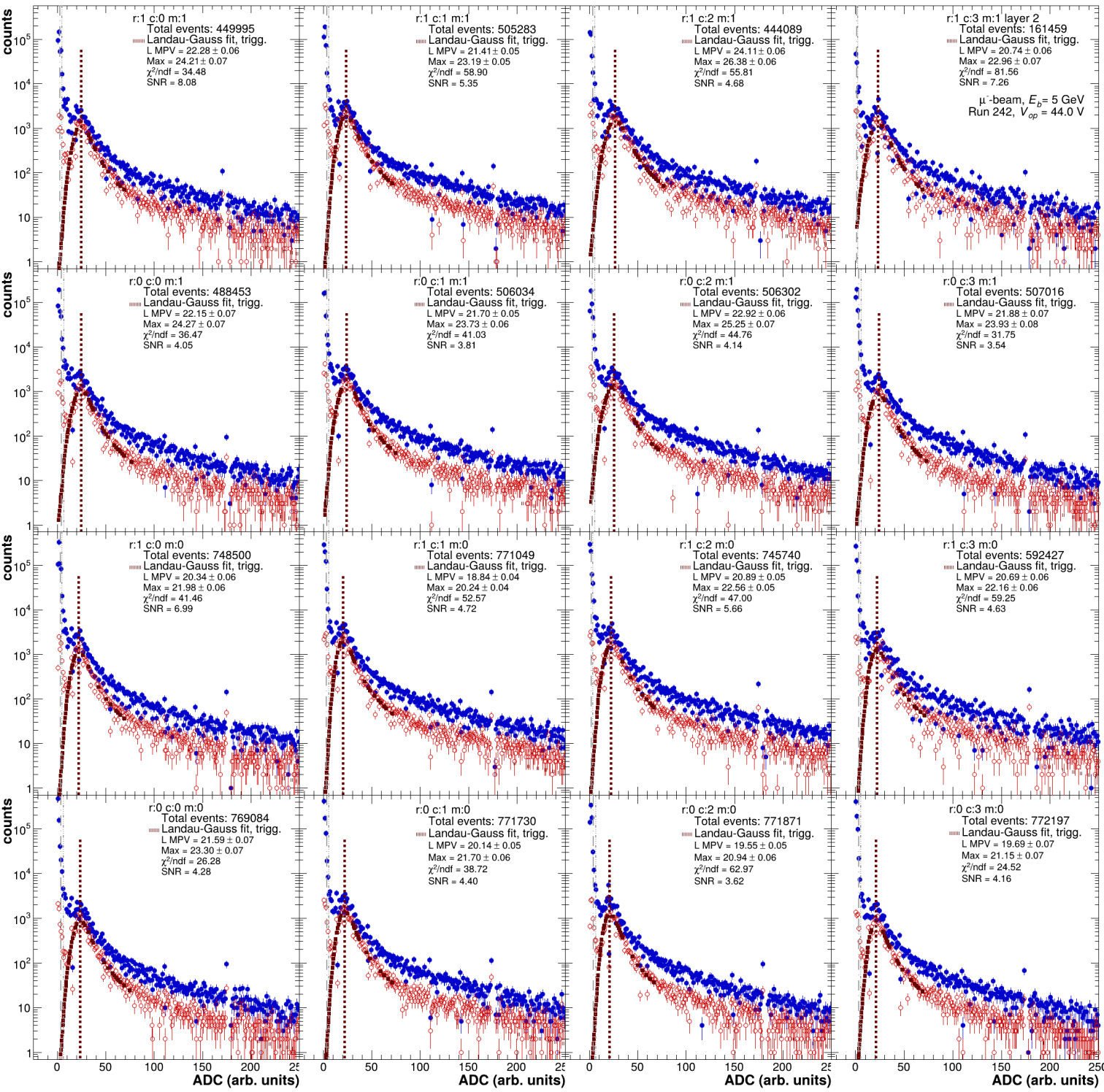

μ⁻-beam, E_b= 5 GeV, Run 242, V_{op} = 44.0 V
brd channel
0
2
4
6
8
10
12
14
0
1
2
3
4
5
6
7
layer
$\sigma_{L,HG,2^{nd}}$ (arb. units)
3
2.8
2.6
2.4
2.2
2
1.8
1.6
1.4

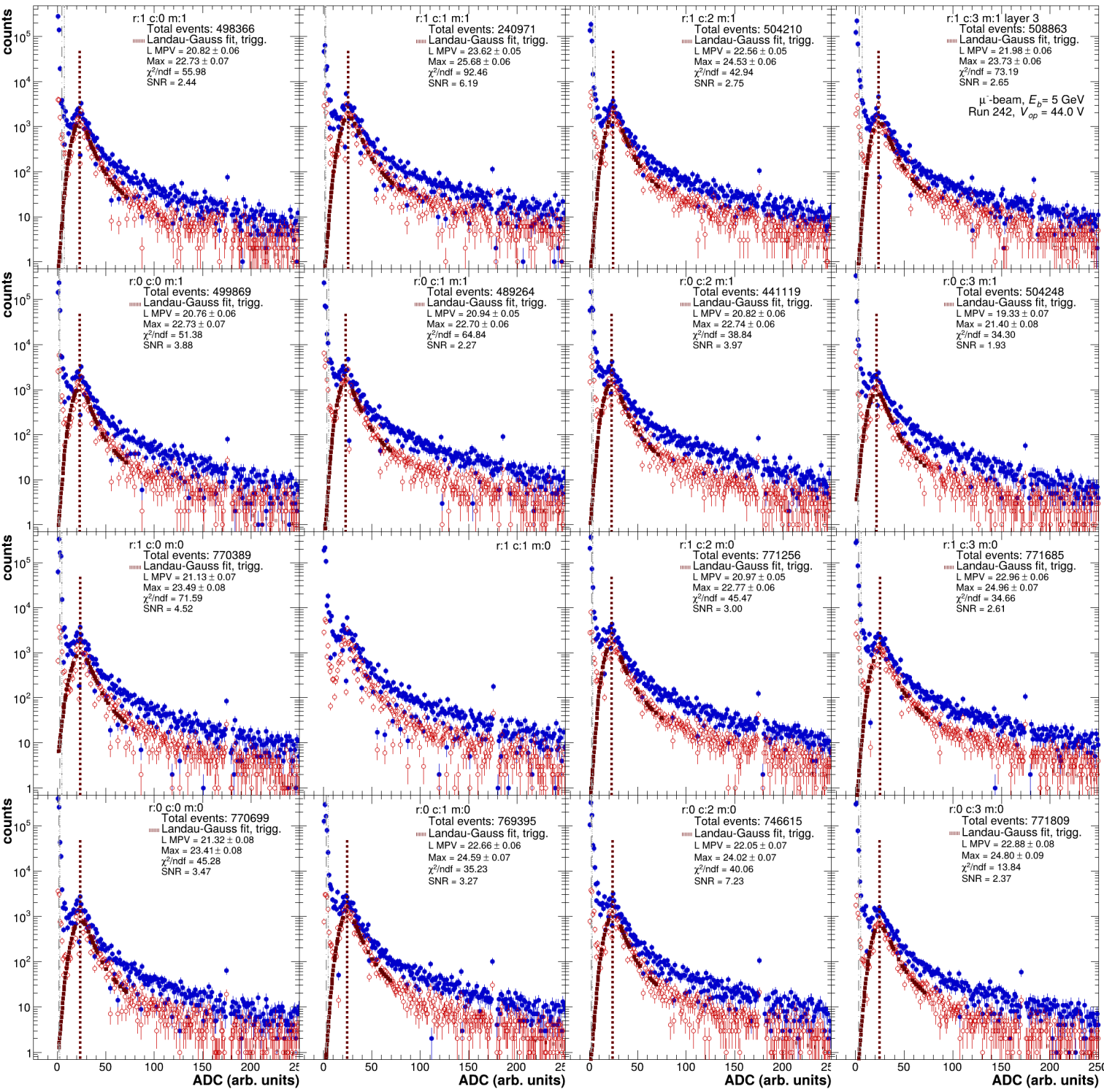

⟨ Max_HG ⟩ = 29.7
μ⁻-beam, E_b = 5 GeV, Run 242, V_op = 44.0 V
brd channel
layer
Max_HG, 2nd (arb. units)
0
2
4
6
8
10
12
14
0
1
2
3
4
5
6
7
15
20
25
30
35
40
45

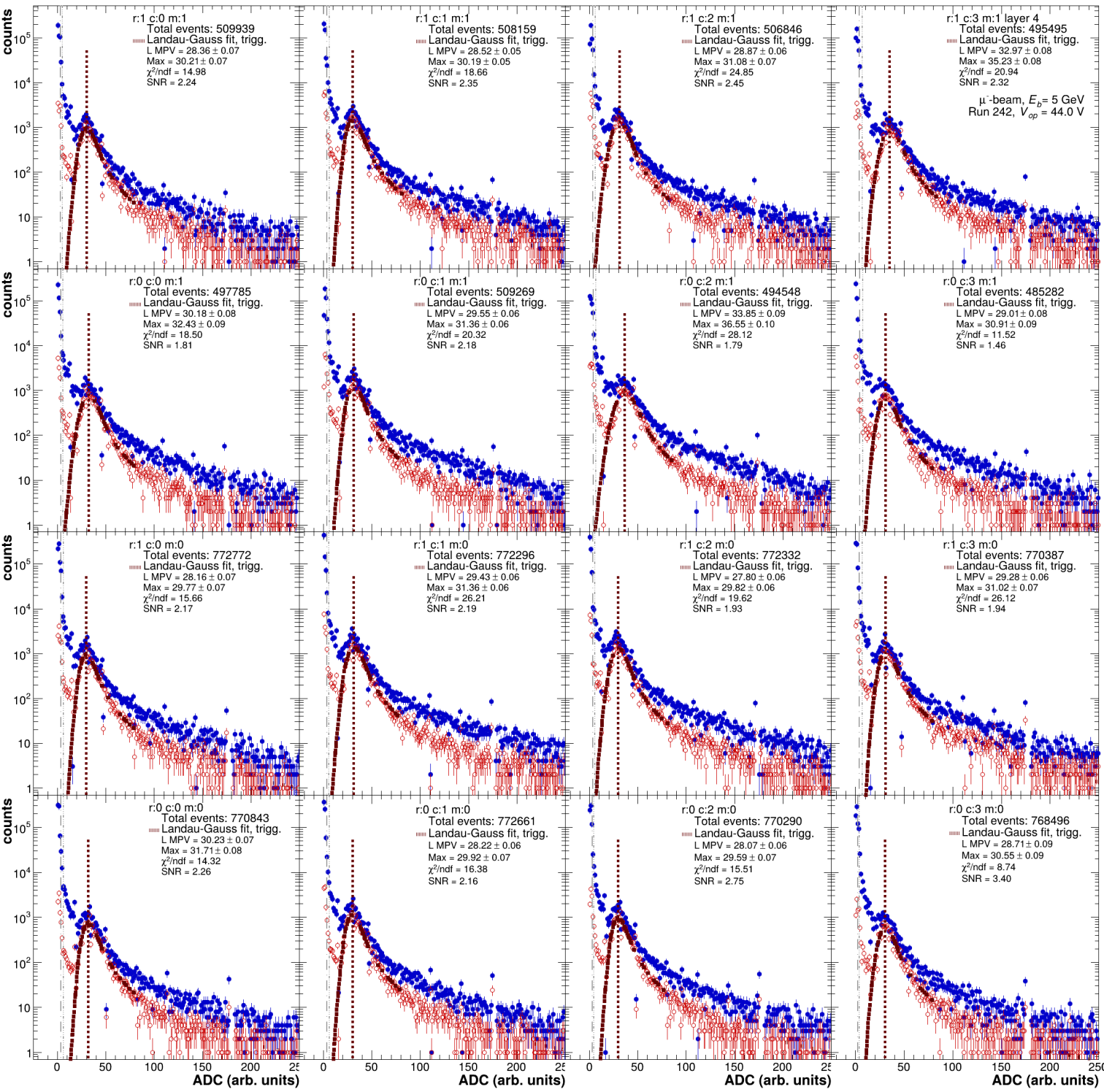

μ⁻-beam, E_b= 5 GeV, Run 242, V_{op} = 44.0 V
brd channel
0
2
4
6
8
10
12
14
0
1
2
3
4
5
6
7
layer
χ^2/ndf
20
40
60
80
100
120
140

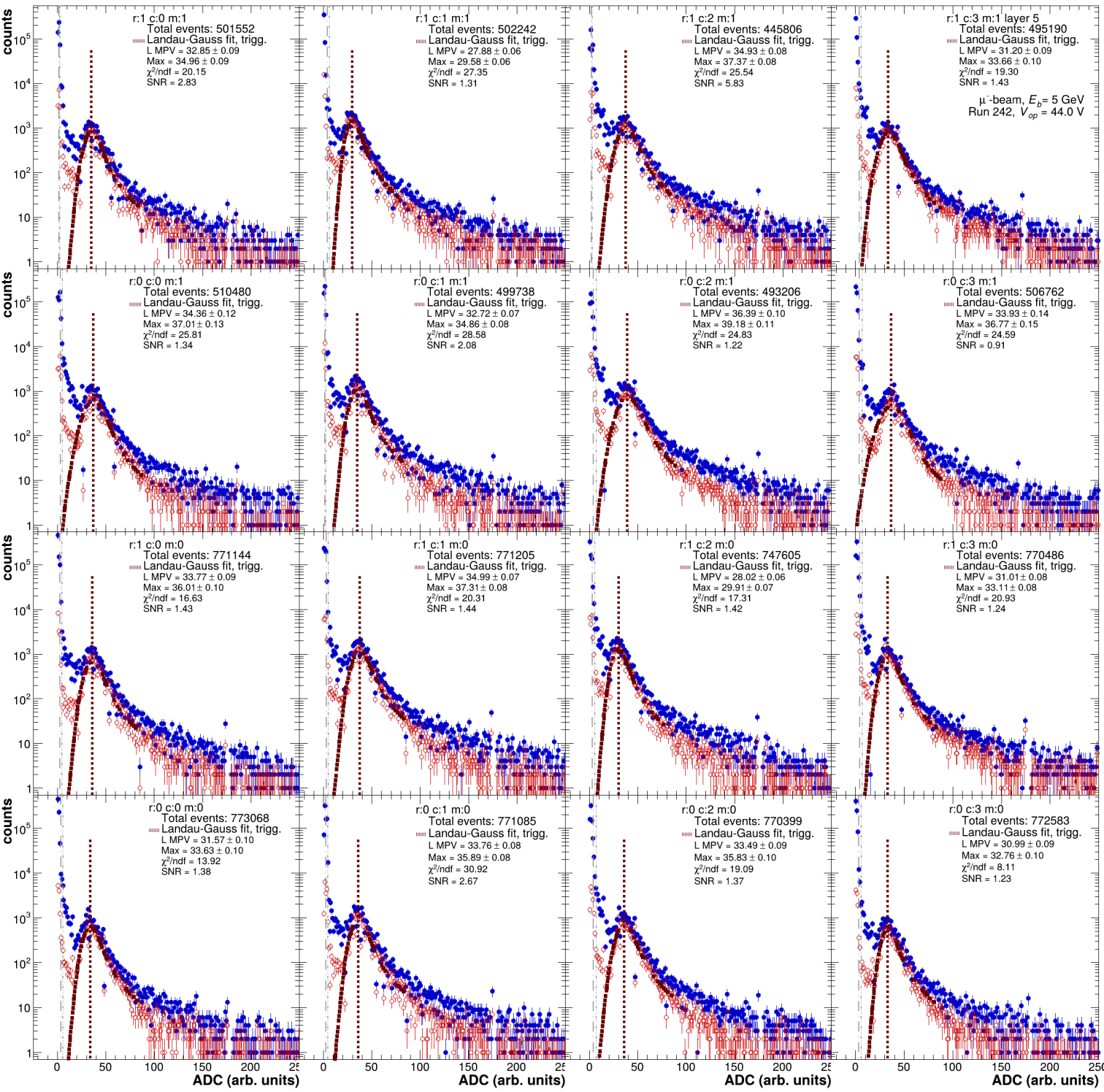

r:1 c:0 m:1
Total events: 489172
Landau-Gauss fit, trigg.
L MPV = 20.94 ± 0.05
Max = 22.63 ± 0.06
χ²/ndf = 40.43
SNR = 5.57
r:1 c:1 m:1
Total events: 449839
Landau-Gauss fit, trigg.
L MPV = 20.74 ± 0.04
Max = 22.32 ± 0.04
χ²/ndf = 64.13
SNR = 7.46
r:1 c:2 m:1
Total events: 400827
Landau-Gauss fit, trigg.
L MPV = 21.46 ± 0.04
Max = 23.11 ± 0.05
χ²/ndf = 77.68
SNR = 5.11
r:1 c:3 m:1 layer 0
Total events: 493764
Landau-Gauss fit, trigg.
L MPV = 21.44 ± 0.05
Max = 23.11 ± 0.05
χ²/ndf = 38.13
SNR = 5.64
μ⁻-beam, E_b = 5 GeV
Run 242, V_op = 44.0 V
r:0 c:0 m:1
Total events: 487697
Landau-Gauss fit, trigg.
L MPV = 21.73 ± 0.06
Max = 23.56 ± 0.07
χ²/ndf = 13.85
SNR = 2.86
r:0 c:1 m:1
Total events: 486577
Landau-Gauss fit, trigg.
L MPV = 20.32 ± 0.05
Max = 22.15 ± 0.05
χ²/ndf = 46.11
SNR = 2.78
r:0 c:2 m:1
Total events: 445707
Landau-Gauss fit, trigg.
L MPV = 20.24 ± 0.05
Max = 22.03 ± 0.06
χ²/ndf = 48.49
SNR = 6.56
r:0 c:3 m:1
Total events: 506290
Landau-Gauss fit, trigg.
L MPV = 19.26 ± 0.06
Max = 21.10 ± 0.07
χ²/ndf = 35.88
SNR = 2.21
r:1 c:0 m:0
Total events: 769539
Landau-Gauss fit, trigg.
L MPV = 15.75 ± 0.04
Max = 16.58 ± 0.04
χ²/ndf = 86.38
SNR = 4.31
r:1 c:1 m:0
Total events: 748610
Landau-Gauss fit, trigg.
L MPV = 15.81 ± 0.03
Max = 16.91 ± 0.03
χ²/ndf = 91.88
SNR = 4.06
r:1 c:2 m:0
Total events: 725871
Landau-Gauss fit, trigg.
L MPV = 17.81 ± 0.04
Max = 19.35 ± 0.04
χ²/ndf = 68.06
SNR = 10.37
r:1 c:3 m:0
Total events: 769895
Landau-Gauss fit, trigg.
L MPV = 14.62 ± 0.03
Max = 14.76 ± 0.03
χ²/ndf = 138.57
SNR = 4.29
r:0 c:0 m:0
Total events: 769337
Landau-Gauss fit, trigg.
L MPV = 15.87 ± 0.05
Max = 17.26 ± 0.06
χ²/ndf = 52.84
SNR = 4.32
r:0 c:1 m:0
Total events: 768784
Landau-Gauss fit, trigg.
L MPV = 15.30 ± 0.04
Max = 16.57 ± 0.04
χ²/ndf = 76.70
SNR = 6.38
r:0 c:2 m:0
Total events: 747423
Landau-Gauss fit, trigg.
L MPV = 16.32 ± 0.04
Max = 17.53 ± 0.05
χ²/ndf = 60.05
SNR = 4.26
r:0 c:3 m:0
Total events: 772483
Landau-Gauss fit, trigg.
L MPV = 14.43 ± 0.05
Max = 15.71 ± 0.06
χ²/ndf = 53.87
SNR = 3.22
counts
ADC (arb. units)

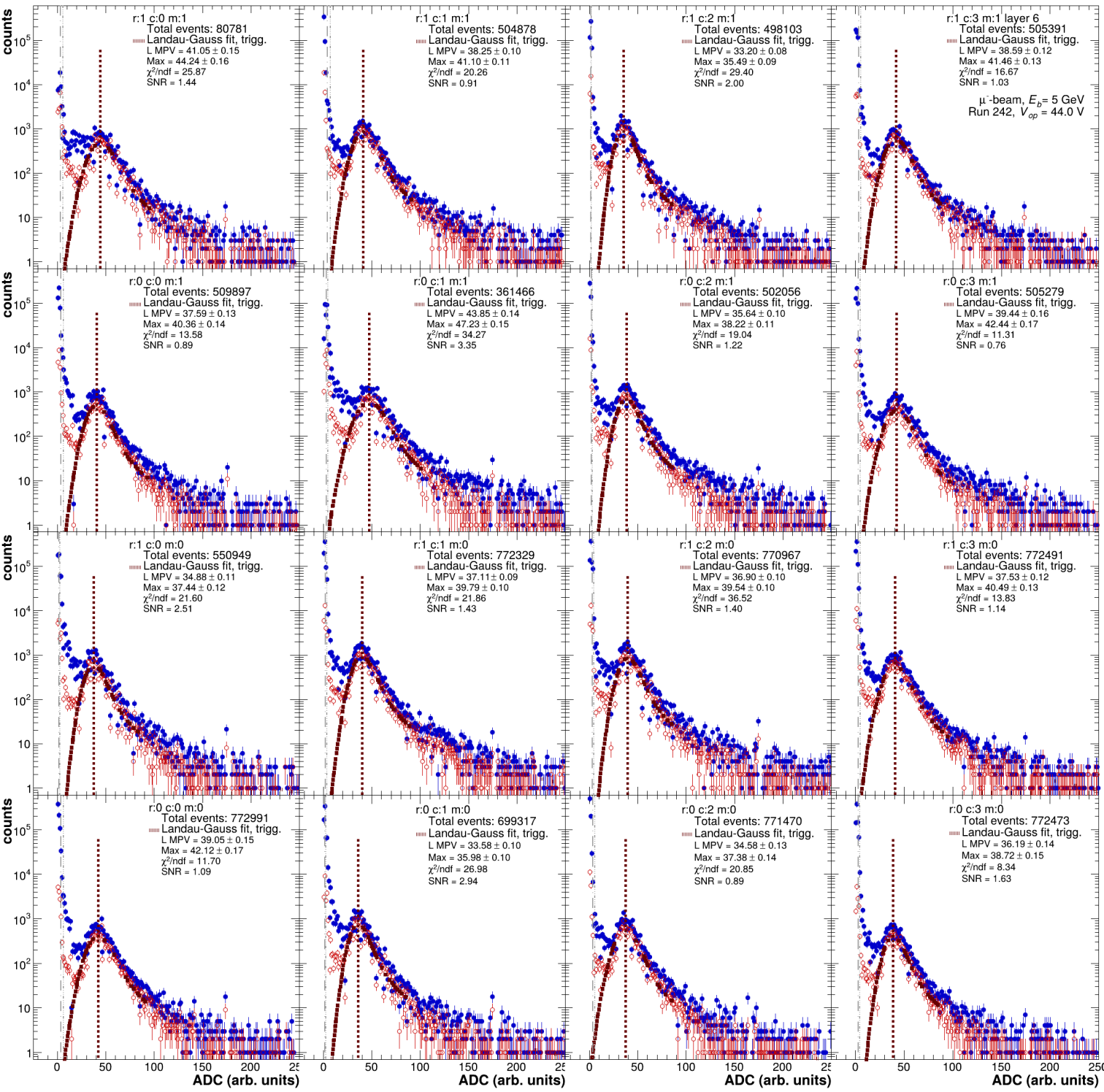

r:1 c:0 m:1
Total events: 487386
Landau-Gauss fit, trigg.
L MPV = 20.75 ± 0.05
Max = 22.44 ± 0.06
χ²/ndf = 44.92
SNR = 7.11
r:1 c:1 m:1
Total events: 488960
Landau-Gauss fit, trigg.
L MPV = 21.96 ± 0.04
Max = 23.69 ± 0.05
χ²/ndf = 45.14
SNR = 7.13
r:1 c:2 m:1
Total events: 490806
Landau-Gauss fit, trigg.
L MPV = 20.90 ± 0.05
Max = 22.58 ± 0.05
χ²/ndf = 54.76
SNR = 5.99
r:1 c:3 m:1 layer 1
Total events: 500074
Landau-Gauss fit, trigg.
L MPV = 21.20 ± 0.05
Max = 22.65 ± 0.05
χ²/ndf = 59.12
SNR = 10.39
μ⁻-beam, E_b = 5 GeV
Run 242, V_op = 44.0 V
r:0 c:0 m:1
Total events: 482615
Landau-Gauss fit, trigg.
L MPV = 21.09 ± 0.06
Max = 22.89 ± 0.07
χ²/ndf = 33.54
SNR = 6.29
r:0 c:1 m:1
Total events: 488830
Landau-Gauss fit, trigg.
L MPV = 23.02 ± 0.06
Max = 25.19 ± 0.06
χ²/ndf = 47.58
SNR = 4.25
r:0 c:2 m:1
Total events: 490450
Landau-Gauss fit, trigg.
L MPV = 20.84 ± 0.06
Max = 22.73 ± 0.06
χ²/ndf = 41.54
SNR = 3.83
r:0 c:3 m:1
Total events: 506875
Landau-Gauss fit, trigg.
L MPV = 21.97 ± 0.07
Max = 23.90 ± 0.08
χ²/ndf = 24.88
SNR = 3.82
r:1 c:0 m:0
Total events: 768760
Landau-Gauss fit, trigg.
L MPV = 16.65 ± 0.04
Max = 17.37 ± 0.04
χ²/ndf = 62.66
SNR = 7.14
r:1 c:1 m:0
Total events: 768683
Landau-Gauss fit, trigg.
L MPV = 16.54 ± 0.03
Max = 17.24 ± 0.03
χ²/ndf = 155.22
SNR = 6.19
r:1 c:2 m:0
Total events: 768893
Landau-Gauss fit, trigg.
L MPV = 15.97 ± 0.03
Max = 16.24 ± 0.03
χ²/ndf = 146.59
SNR = 5.48
r:1 c:3 m:0
Total events: 770287
Landau-Gauss fit, trigg.
L MPV = 15.33 ± 0.04
Max = 16.53 ± 0.04
χ²/ndf = 39.54
SNR = 6.09
r:0 c:0 m:0
Total events: 767702
Landau-Gauss fit, trigg.
L MPV = 17.23 ± 0.05
Max = 18.70 ± 0.06
χ²/ndf = 37.08
SNR = 5.37
r:0 c:1 m:0
Total events: 768962
Landau-Gauss fit, trigg.
L MPV = 20.01 ± 0.05
Max = 21.63 ± 0.05
χ²/ndf = 66.13
SNR = 6.91
r:0 c:2 m:0
Total events: 769251
Landau-Gauss fit, trigg.
L MPV = 18.67 ± 0.05
Max = 20.03 ± 0.05
χ²/ndf = 67.81
SNR = 10.83
r:0 c:3 m:0
Total events: 772351
Landau-Gauss fit, trigg.
L MPV = 16.28 ± 0.05
Max = 16.98 ± 0.05
χ²/ndf = 49.53
SNR = 5.60
counts
ADC (arb. units)

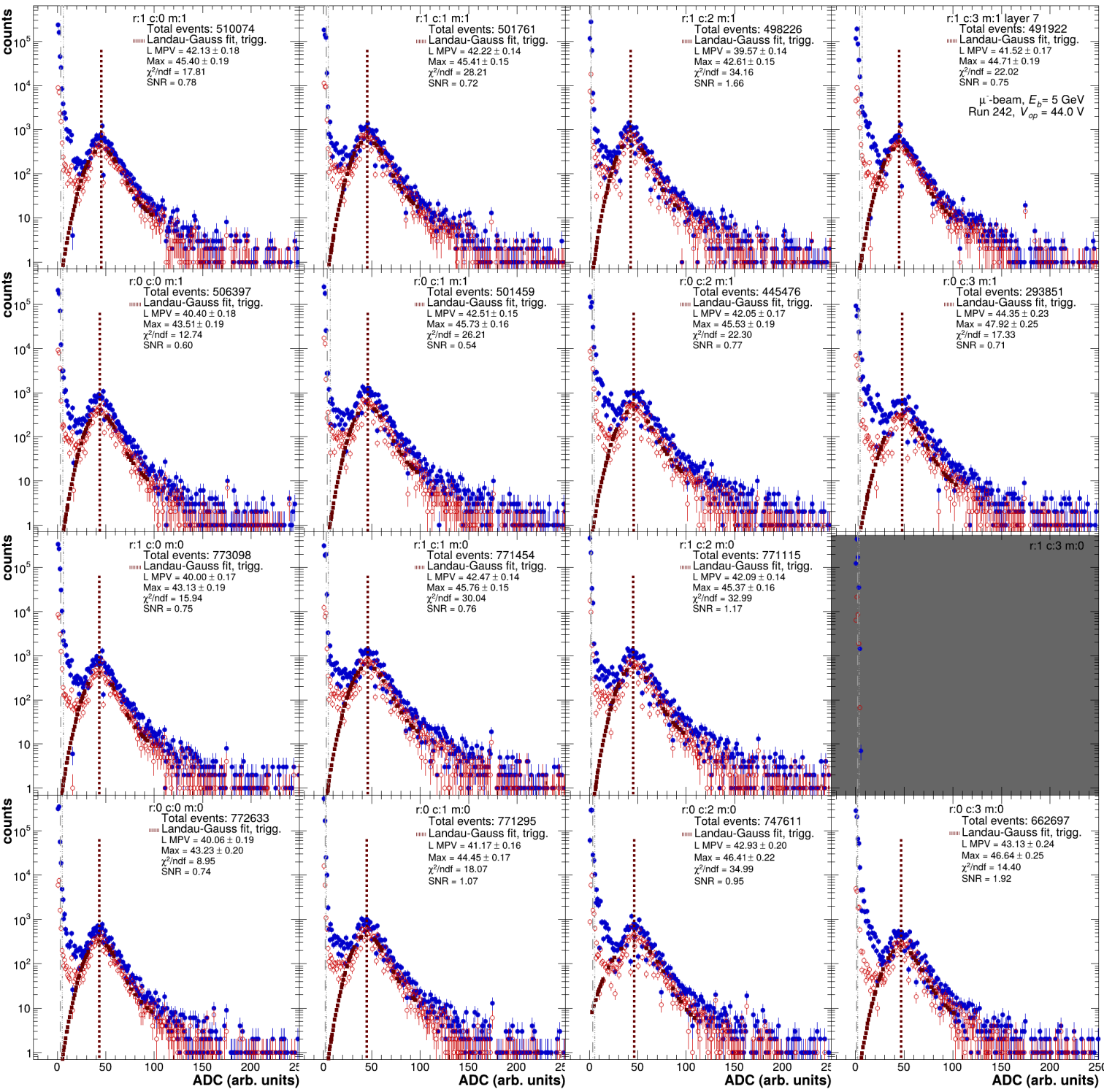

r:1 c:0 m:1
Total events: 449995
Landau-Gauss fit, trigg.
L MPV = 22.28 ± 0.06
Max = 24.21 ± 0.07
χ²/ndf = 34.48
SNR = 8.08
r:1 c:1 m:1
Total events: 505283
Landau-Gauss fit, trigg.
L MPV = 21.41 ± 0.05
Max = 23.19 ± 0.05
χ²/ndf = 58.90
SNR = 5.35
r:1 c:2 m:1
Total events: 444089
Landau-Gauss fit, trigg.
L MPV = 24.11 ± 0.06
Max = 26.38 ± 0.06
χ²/ndf = 55.81
SNR = 4.68
r:1 c:3 m:1 layer 2
Total events: 161459
Landau-Gauss fit, trigg.
L MPV = 20.74 ± 0.06
Max = 22.96 ± 0.07
χ²/ndf = 81.56
SNR = 7.26
μ⁻-beam, E_b = 5 GeV
Run 242, V_op = 44.0 V
r:0 c:0 m:1
Total events: 488453
Landau-Gauss fit, trigg.
L MPV = 22.15 ± 0.07
Max = 24.27 ± 0.07
χ²/ndf = 36.47
SNR = 4.05
r:0 c:1 m:1
Total events: 506034
Landau-Gauss fit, trigg.
L MPV = 21.70 ± 0.05
Max = 23.73 ± 0.06
χ²/ndf = 41.03
SNR = 3.81
r:0 c:2 m:1
Total events: 506302
Landau-Gauss fit, trigg.
L MPV = 22.92 ± 0.06
Max = 25.25 ± 0.07
χ²/ndf = 44.76
SNR = 4.14
r:0 c:3 m:1
Total events: 507016
Landau-Gauss fit, trigg.
L MPV = 21.88 ± 0.07
Max = 23.93 ± 0.08
χ²/ndf = 31.75
SNR = 3.54
r:1 c:0 m:0
Total events: 748500
Landau-Gauss fit, trigg.
L MPV = 20.34 ± 0.06
Max = 21.98 ± 0.06
χ²/ndf = 41.46
SNR = 6.99
r:1 c:1 m:0
Total events: 771049
Landau-Gauss fit, trigg.
L MPV = 18.84 ± 0.04
Max = 20.24 ± 0.04
χ²/ndf = 52.57
SNR = 4.72
r:1 c:2 m:0
Total events: 745740
Landau-Gauss fit, trigg.
L MPV = 20.89 ± 0.05
Max = 22.56 ± 0.05
χ²/ndf = 47.00
SNR = 5.66
r:1 c:3 m:0
Total events: 592427
Landau-Gauss fit, trigg.
L MPV = 20.69 ± 0.06
Max = 22.16 ± 0.06
χ²/ndf = 59.25
SNR = 4.63
r:0 c:0 m:0
Total events: 769084
Landau-Gauss fit, trigg.
L MPV = 21.59 ± 0.07
Max = 23.30 ± 0.07
χ²/ndf = 26.28
SNR = 4.28
r:0 c:1 m:0
Total events: 771730
Landau-Gauss fit, trigg.
L MPV = 20.14 ± 0.05
Max = 21.70 ± 0.06
χ²/ndf = 38.72
SNR = 4.40
r:0 c:2 m:0
Total events: 771871
Landau-Gauss fit, trigg.
L MPV = 19.55 ± 0.05
Max = 20.94 ± 0.06
χ²/ndf = 62.97
SNR = 3.62
r:0 c:3 m:0
Total events: 772197
Landau-Gauss fit, trigg.
L MPV = 19.69 ± 0.07
Max = 21.15 ± 0.07
χ²/ndf = 24.52
SNR = 4.16
counts
ADC (arb. units)

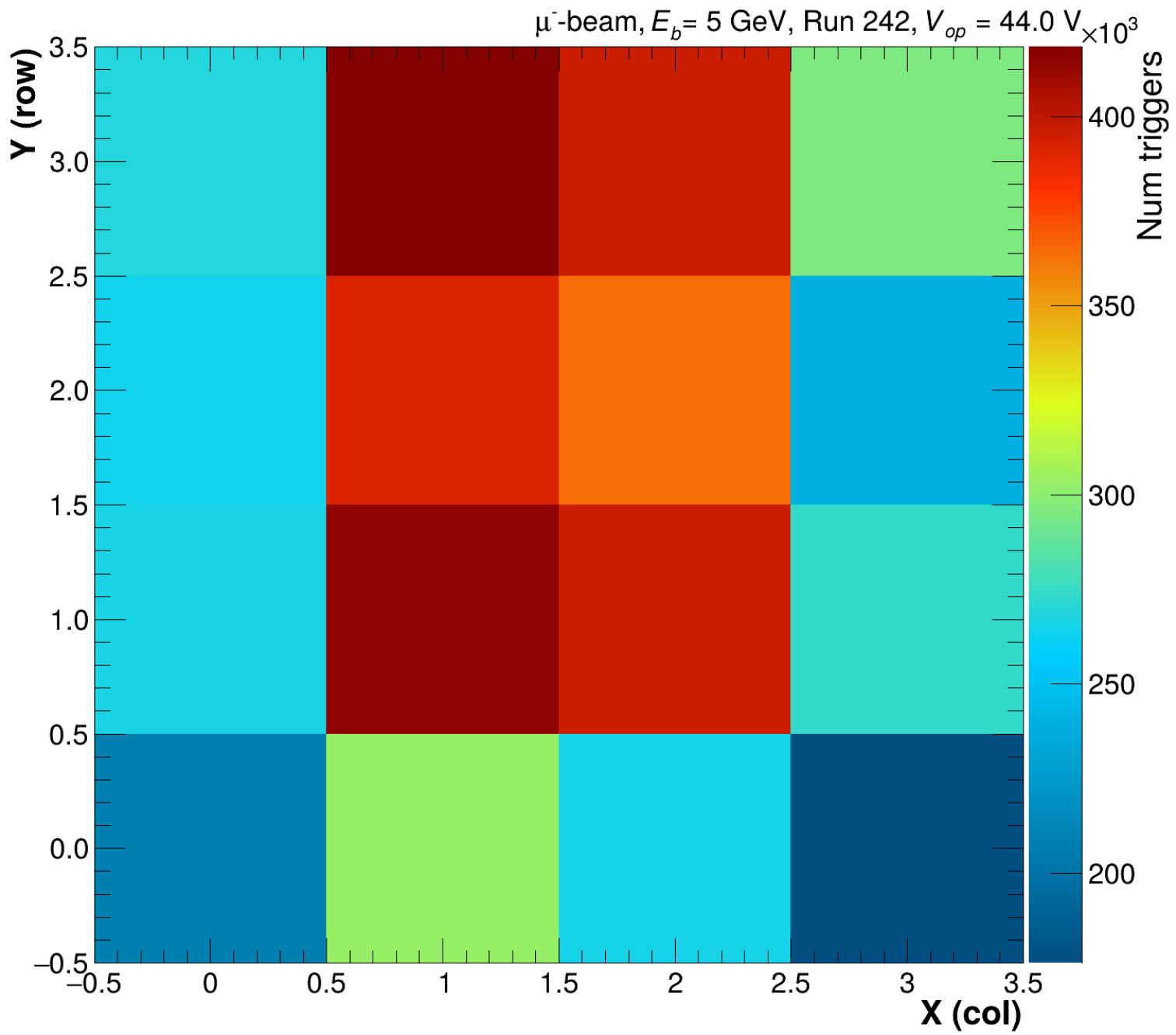

r:1 c:0 m:1
Total events: 498366
Landau-Gauss fit, trigg.
L MPV = 20.82 ± 0.06
Max = 22.73 ± 0.07
χ²/ndf = 55.98
SNR = 2.44
r:1 c:1 m:1
Total events: 240971
Landau-Gauss fit, trigg.
L MPV = 23.62 ± 0.05
Max = 25.68 ± 0.06
χ²/ndf = 92.46
SNR = 6.19
r:1 c:2 m:1
Total events: 504210
Landau-Gauss fit, trigg.
L MPV = 22.56 ± 0.05
Max = 24.53 ± 0.06
χ²/ndf = 42.94
SNR = 2.75
r:1 c:3 m:1 layer 3
Total events: 508863
Landau-Gauss fit, trigg.
L MPV = 21.98 ± 0.06
Max = 23.73 ± 0.06
χ²/ndf = 73.19
SNR = 2.65
μ⁻-beam, E_b = 5 GeV
Run 242, V_{op} = 44.0 V
r:0 c:0 m:1
Total events: 499869
Landau-Gauss fit, trigg.
L MPV = 20.76 ± 0.06
Max = 22.73 ± 0.07
χ²/ndf = 51.38
SNR = 3.88
r:0 c:1 m:1
Total events: 489264
Landau-Gauss fit, trigg.
L MPV = 20.94 ± 0.05
Max = 22.70 ± 0.06
χ²/ndf = 64.84
SNR = 2.27
r:0 c:2 m:1
Total events: 441119
Landau-Gauss fit, trigg.
L MPV = 20.82 ± 0.06
Max = 22.74 ± 0.06
χ²/ndf = 38.84
SNR = 3.97
r:0 c:3 m:1
Total events: 504248
Landau-Gauss fit, trigg.
L MPV = 19.33 ± 0.07
Max = 21.40 ± 0.08
χ²/ndf = 34.30
SNR = 1.93
r:1 c:0 m:0
Total events: 770389
Landau-Gauss fit, trigg.
L MPV = 21.13 ± 0.07
Max = 23.49 ± 0.08
χ²/ndf = 71.59
SNR = 4.52
r:1 c:1 m:0
r:1 c:2 m:0
Total events: 771256
Landau-Gauss fit, trigg.
L MPV = 20.97 ± 0.05
Max = 22.77 ± 0.06
χ²/ndf = 45.47
SNR = 3.00
r:1 c:3 m:0
Total events: 771685
Landau-Gauss fit, trigg.
L MPV = 22.96 ± 0.06
Max = 24.96 ± 0.07
χ²/ndf = 34.66
SNR = 2.61
r:0 c:0 m:0
Total events: 770699
Landau-Gauss fit, trigg.
L MPV = 21.32 ± 0.08
Max = 23.41 ± 0.08
χ²/ndf = 45.28
SNR = 3.47
r:0 c:1 m:0
Total events: 769395
Landau-Gauss fit, trigg.
L MPV = 22.66 ± 0.06
Max = 24.59 ± 0.07
χ²/ndf = 35.23
SNR = 3.27
r:0 c:2 m:0
Total events: 746615
Landau-Gauss fit, trigg.
L MPV = 22.05 ± 0.07
Max = 24.02 ± 0.07
χ²/ndf = 40.06
SNR = 7.23
r:0 c:3 m:0
Total events: 771809
Landau-Gauss fit, trigg.
L MPV = 22.88 ± 0.08
Max = 24.80 ± 0.09
χ²/ndf = 13.84
SNR = 2.37
counts
ADC (arb. units)

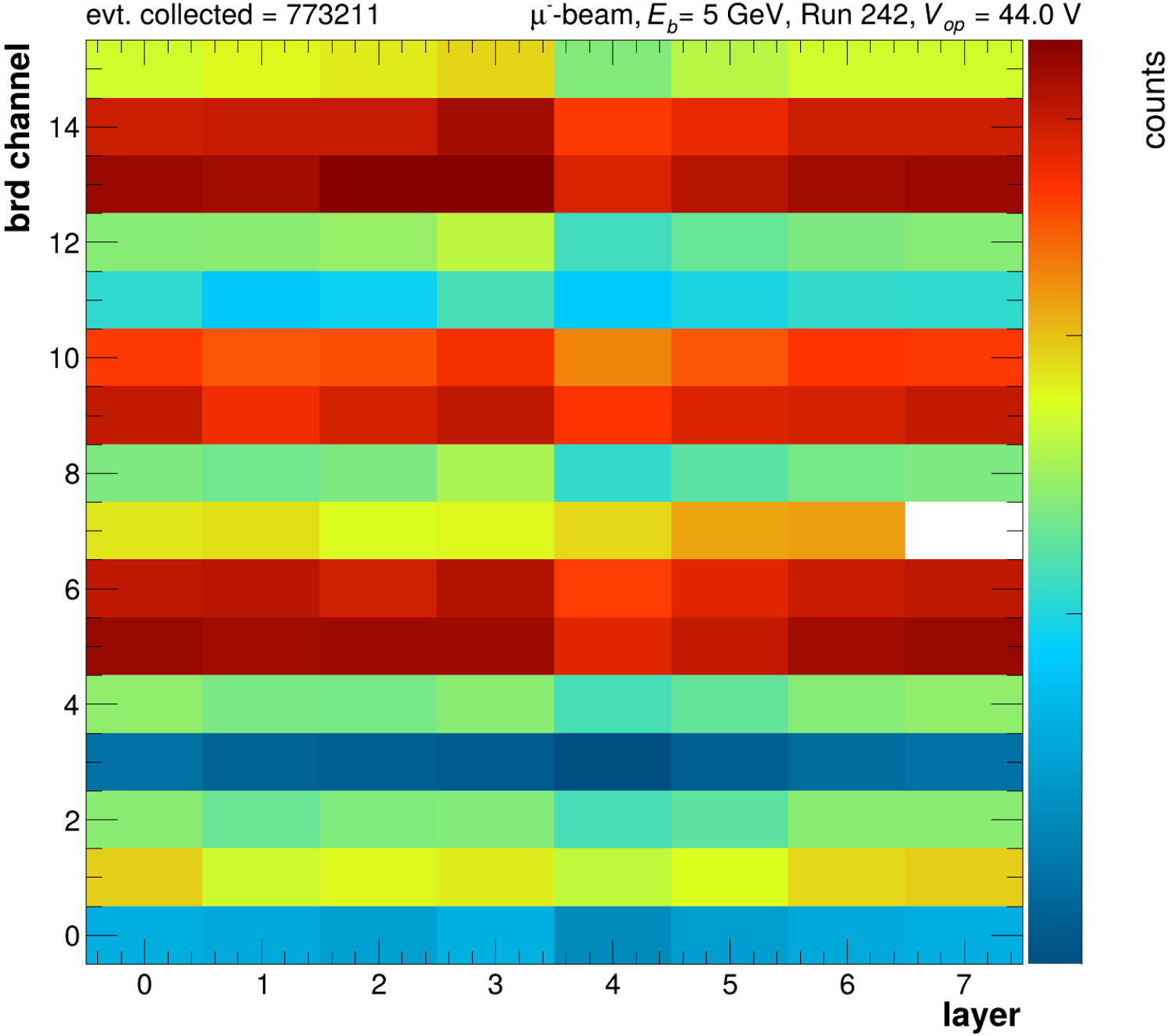

r:1 c:0 m:1
Total events: 509939
Landau-Gauss fit, trigg.
L MPV = 28.36 ± 0.07
Max = 30.21 ± 0.07
χ²/ndf = 14.98
SNR = 2.24
r:1 c:1 m:1
Total events: 508159
Landau-Gauss fit, trigg.
L MPV = 28.52 ± 0.05
Max = 30.19 ± 0.05
χ²/ndf = 18.66
SNR = 2.35
r:1 c:2 m:1
Total events: 506846
Landau-Gauss fit, trigg.
L MPV = 28.87 ± 0.06
Max = 31.08 ± 0.07
χ²/ndf = 24.85
SNR = 2.45
r:1 c:3 m:1 layer 4
Total events: 495495
Landau-Gauss fit, trigg.
L MPV = 32.97 ± 0.08
Max = 35.23 ± 0.08
χ²/ndf = 20.94
SNR = 2.32
μ⁻-beam, E_b = 5 GeV
Run 242, V_op = 44.0 V
r:0 c:0 m:1
Total events: 497785
Landau-Gauss fit, trigg.
L MPV = 30.18 ± 0.08
Max = 32.43 ± 0.09
χ²/ndf = 18.50
SNR = 1.81
r:0 c:1 m:1
Total events: 509269
Landau-Gauss fit, trigg.
L MPV = 29.55 ± 0.06
Max = 31.36 ± 0.06
χ²/ndf = 20.32
SNR = 2.18
r:0 c:2 m:1
Total events: 494548
Landau-Gauss fit, trigg.
L MPV = 33.85 ± 0.09
Max = 36.55 ± 0.10
χ²/ndf = 28.12
SNR = 1.79
r:0 c:3 m:1
Total events: 485282
Landau-Gauss fit, trigg.
L MPV = 29.01 ± 0.08
Max = 30.91 ± 0.09
χ²/ndf = 11.52
SNR = 1.46
r:1 c:0 m:0
Total events: 772772
Landau-Gauss fit, trigg.
L MPV = 28.16 ± 0.07
Max = 29.77 ± 0.07
χ²/ndf = 15.66
SNR = 2.17
r:1 c:1 m:0
Total events: 772296
Landau-Gauss fit, trigg.
L MPV = 29.43 ± 0.06
Max = 31.36 ± 0.06
χ²/ndf = 26.21
SNR = 2.19
r:1 c:2 m:0
Total events: 772332
Landau-Gauss fit, trigg.
L MPV = 27.80 ± 0.06
Max = 29.82 ± 0.06
χ²/ndf = 19.62
SNR = 1.93
r:1 c:3 m:0
Total events: 770387
Landau-Gauss fit, trigg.
L MPV = 29.28 ± 0.06
Max = 31.02 ± 0.07
χ²/ndf = 26.12
SNR = 1.94
r:0 c:0 m:0
Total events: 770843
Landau-Gauss fit, trigg.
L MPV = 30.23 ± 0.07
Max = 31.71 ± 0.08
χ²/ndf = 14.32
SNR = 2.26
r:0 c:1 m:0
Total events: 772661
Landau-Gauss fit, trigg.
L MPV = 28.22 ± 0.06
Max = 29.92 ± 0.07
χ²/ndf = 16.38
SNR = 2.16
r:0 c:2 m:0
Total events: 770290
Landau-Gauss fit, trigg.
L MPV = 28.07 ± 0.06
Max = 29.59 ± 0.07
χ²/ndf = 15.51
SNR = 2.75
r:0 c:3 m:0
Total events: 768496
Landau-Gauss fit, trigg.
L MPV = 28.71 ± 0.09
Max = 30.55 ± 0.09
χ²/ndf = 8.74
SNR = 3.40
counts
ADC (arb. units)

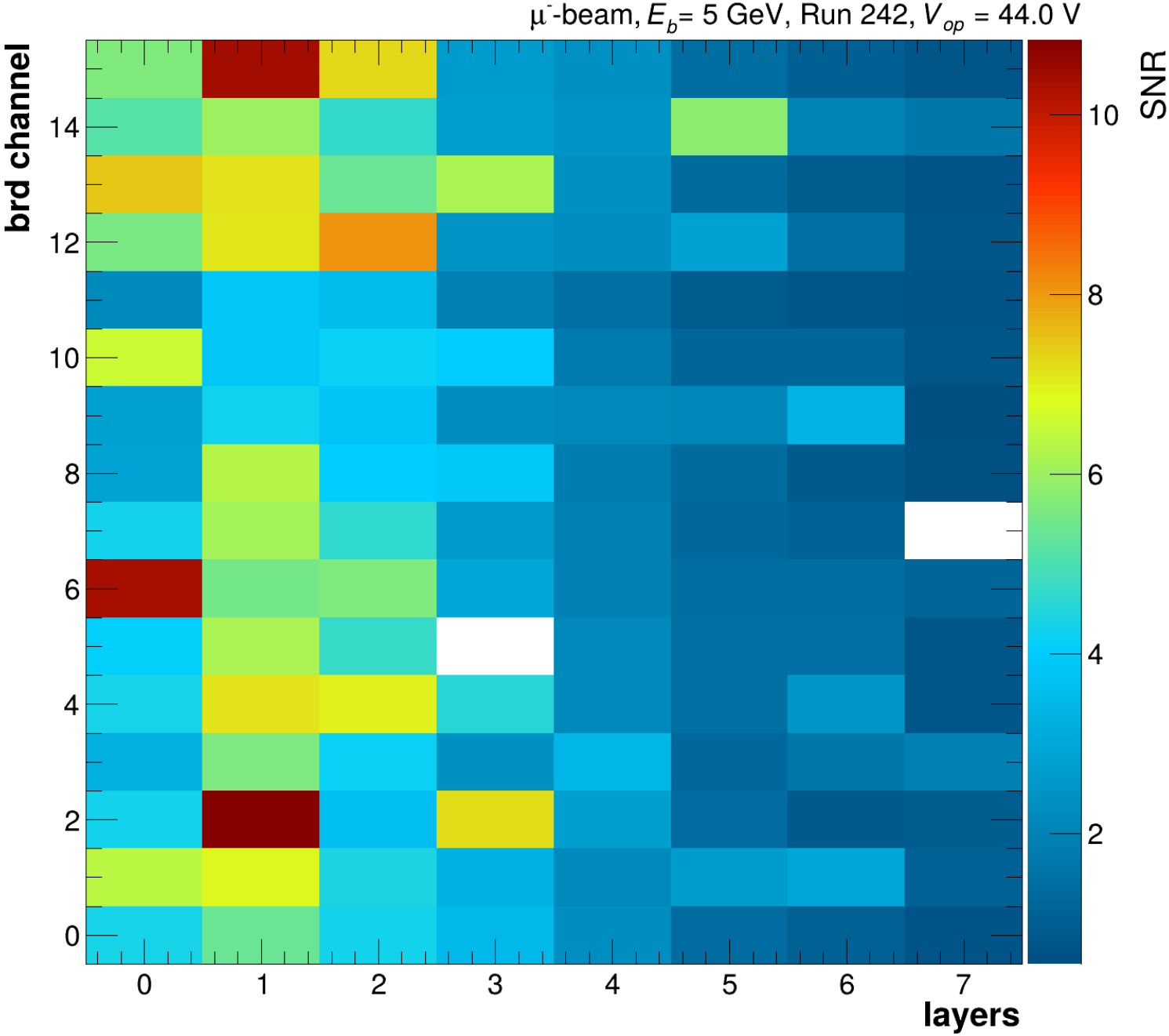

r:1 c:0 m:1
Total events: 501552
Landau-Gauss fit, trigg.
L MPV = 32.85 ± 0.09
Max = 34.96 ± 0.09
χ²/ndf = 20.15
SNR = 2.83
r:1 c:1 m:1
Total events: 502242
Landau-Gauss fit, trigg.
L MPV = 27.88 ± 0.06
Max = 29.58 ± 0.06
χ²/ndf = 27.35
SNR = 1.31
r:1 c:2 m:1
Total events: 445806
Landau-Gauss fit, trigg.
L MPV = 34.93 ± 0.08
Max = 37.37 ± 0.08
χ²/ndf = 25.54
SNR = 5.83
r:1 c:3 m:1 layer 5
Total events: 495190
Landau-Gauss fit, trigg.
L MPV = 31.20 ± 0.09
Max = 33.66 ± 0.10
χ²/ndf = 19.30
SNR = 1.43
μ⁻-beam, E_b = 5 GeV
Run 242, V_{op} = 44.0 V
r:0 c:0 m:1
Total events: 510480
Landau-Gauss fit, trigg.
L MPV = 34.36 ± 0.12
Max = 37.01 ± 0.13
χ²/ndf = 25.81
SNR = 1.34
r:0 c:1 m:1
Total events: 499738
Landau-Gauss fit, trigg.
L MPV = 32.72 ± 0.07
Max = 34.86 ± 0.08
χ²/ndf = 28.58
SNR = 2.08
r:0 c:2 m:1
Total events: 493206
Landau-Gauss fit, trigg.
L MPV = 36.39 ± 0.10
Max = 39.18 ± 0.11
χ²/ndf = 24.83
SNR = 1.22
r:0 c:3 m:1
Total events: 506762
Landau-Gauss fit, trigg.
L MPV = 33.93 ± 0.14
Max = 36.77 ± 0.15
χ²/ndf = 24.59
SNR = 0.91
r:1 c:0 m:0
Total events: 771144
Landau-Gauss fit, trigg.
L MPV = 33.77 ± 0.09
Max = 36.01 ± 0.10
χ²/ndf = 16.63
SNR = 1.43
r:1 c:1 m:0
Total events: 771205
Landau-Gauss fit, trigg.
L MPV = 34.99 ± 0.07
Max = 37.31 ± 0.08
χ²/ndf = 20.31
SNR = 1.44
r:1 c:2 m:0
Total events: 747605
Landau-Gauss fit, trigg.
L MPV = 28.02 ± 0.06
Max = 29.91 ± 0.07
χ²/ndf = 17.31
SNR = 1.42
r:1 c:3 m:0
Total events: 770486
Landau-Gauss fit, trigg.
L MPV = 31.01 ± 0.08
Max = 33.11 ± 0.08
χ²/ndf = 20.93
SNR = 1.24
r:0 c:0 m:0
Total events: 773068
Landau-Gauss fit, trigg.
L MPV = 31.57 ± 0.10
Max = 33.63 ± 0.10
χ²/ndf = 13.92
SNR = 1.38
r:0 c:1 m:0
Total events: 771085
Landau-Gauss fit, trigg.
L MPV = 33.76 ± 0.08
Max = 35.89 ± 0.08
χ²/ndf = 30.92
SNR = 2.67
r:0 c:2 m:0
Total events: 770399
Landau-Gauss fit, trigg.
L MPV = 33.49 ± 0.09
Max = 35.83 ± 0.10
χ²/ndf = 19.09
SNR = 1.37
r:0 c:3 m:0
Total events: 772583
Landau-Gauss fit, trigg.
L MPV = 30.99 ± 0.09
Max = 32.76 ± 0.10
χ²/ndf = 8.11
SNR = 1.23
counts
ADC (arb. units)

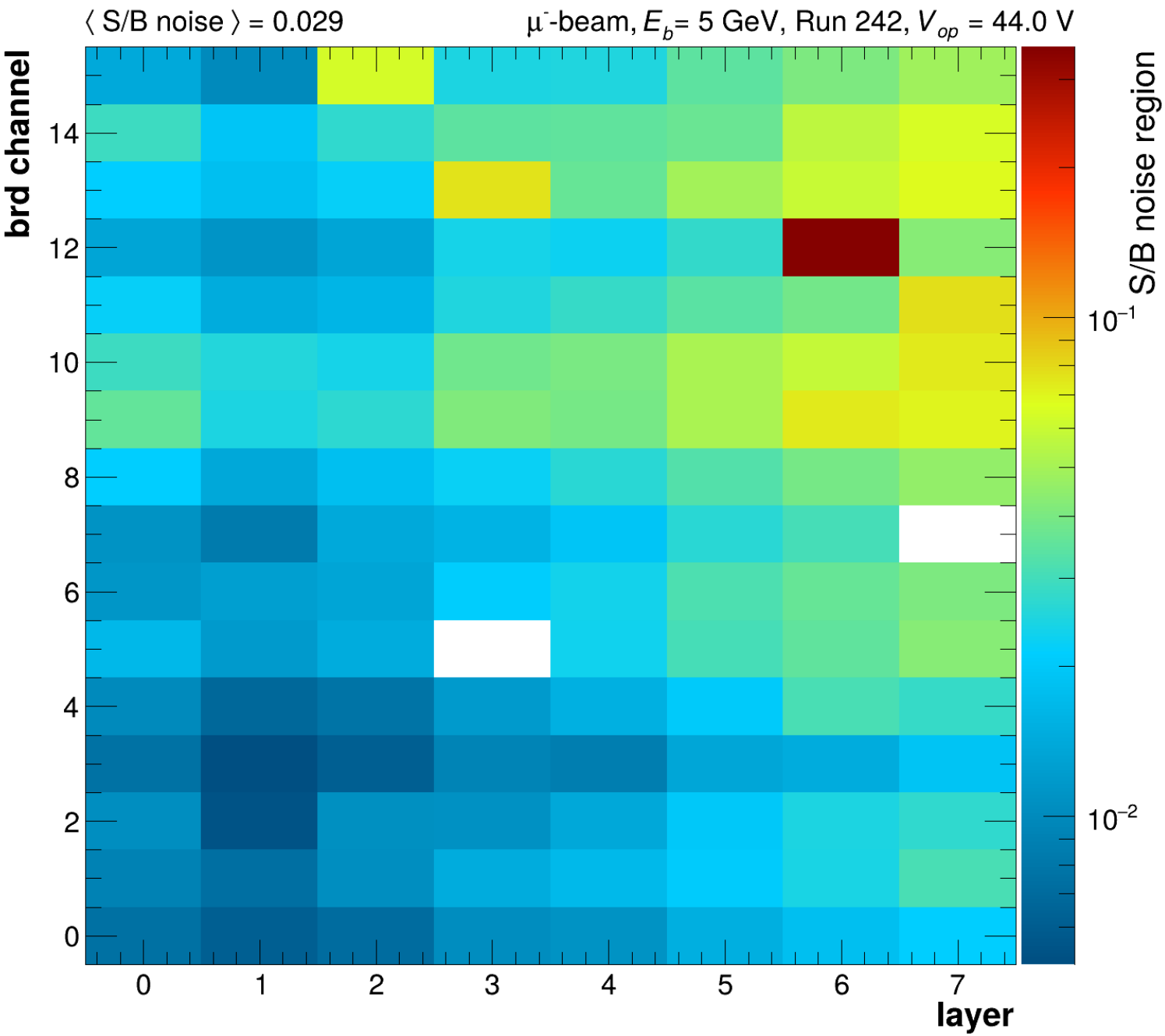

r:1 c:0 m:1
Total events: 80781
Landau-Gauss fit, trigg.
L MPV = 41.05 ± 0.15
Max = 44.24 ± 0.16
χ²/ndf = 25.87
SNR = 1.44
r:1 c:1 m:1
Total events: 504878
Landau-Gauss fit, trigg.
L MPV = 38.25 ± 0.10
Max = 41.10 ± 0.11
χ²/ndf = 20.26
SNR = 0.91
r:1 c:2 m:1
Total events: 498103
Landau-Gauss fit, trigg.
L MPV = 33.20 ± 0.08
Max = 35.49 ± 0.09
χ²/ndf = 29.40
SNR = 2.00
r:1 c:3 m:1 layer 6
Total events: 505391
Landau-Gauss fit, trigg.
L MPV = 38.59 ± 0.12
Max = 41.46 ± 0.13
χ²/ndf = 16.67
SNR = 1.03
μ⁻-beam, E_b = 5 GeV
Run 242, V_{op} = 44.0 V
r:0 c:0 m:1
Total events: 509897
Landau-Gauss fit, trigg.
L MPV = 37.59 ± 0.13
Max = 40.36 ± 0.14
χ²/ndf = 13.58
SNR = 0.89
r:0 c:1 m:1
Total events: 361466
Landau-Gauss fit, trigg.
L MPV = 43.85 ± 0.14
Max = 47.23 ± 0.15
χ²/ndf = 34.27
SNR = 3.35
r:0 c:2 m:1
Total events: 502056
Landau-Gauss fit, trigg.
L MPV = 35.64 ± 0.10
Max = 38.22 ± 0.11
χ²/ndf = 19.04
SNR = 1.22
r:0 c:3 m:1
Total events: 505279
Landau-Gauss fit, trigg.
L MPV = 39.44 ± 0.16
Max = 42.44 ± 0.17
χ²/ndf = 11.31
SNR = 0.76
r:1 c:0 m:0
Total events: 550949
Landau-Gauss fit, trigg.
L MPV = 34.88 ± 0.11
Max = 37.44 ± 0.12
χ²/ndf = 21.60
SNR = 2.51
r:1 c:1 m:0
Total events: 772329
Landau-Gauss fit, trigg.
L MPV = 37.11 ± 0.09
Max = 39.79 ± 0.10
χ²/ndf = 21.86
SNR = 1.43
r:1 c:2 m:0
Total events: 770967
Landau-Gauss fit, trigg.
L MPV = 36.90 ± 0.10
Max = 39.54 ± 0.10
χ²/ndf = 36.52
SNR = 1.40
r:1 c:3 m:0
Total events: 772491
Landau-Gauss fit, trigg.
L MPV = 37.53 ± 0.12
Max = 40.49 ± 0.13
χ²/ndf = 13.83
SNR = 1.14
r:0 c:0 m:0
Total events: 772991
Landau-Gauss fit, trigg.
L MPV = 39.05 ± 0.15
Max = 42.12 ± 0.17
χ²/ndf = 11.70
SNR = 1.09
r:0 c:1 m:0
Total events: 699317
Landau-Gauss fit, trigg.
L MPV = 33.58 ± 0.10
Max = 35.98 ± 0.10
χ²/ndf = 26.98
SNR = 2.94
r:0 c:2 m:0
Total events: 771470
Landau-Gauss fit, trigg.
L MPV = 34.58 ± 0.13
Max = 37.38 ± 0.14
χ²/ndf = 20.85
SNR = 0.89
r:0 c:3 m:0
Total events: 772473
Landau-Gauss fit, trigg.
L MPV = 36.19 ± 0.14
Max = 38.72 ± 0.15
χ²/ndf = 8.34
SNR = 1.63
counts
ADC (arb. units)

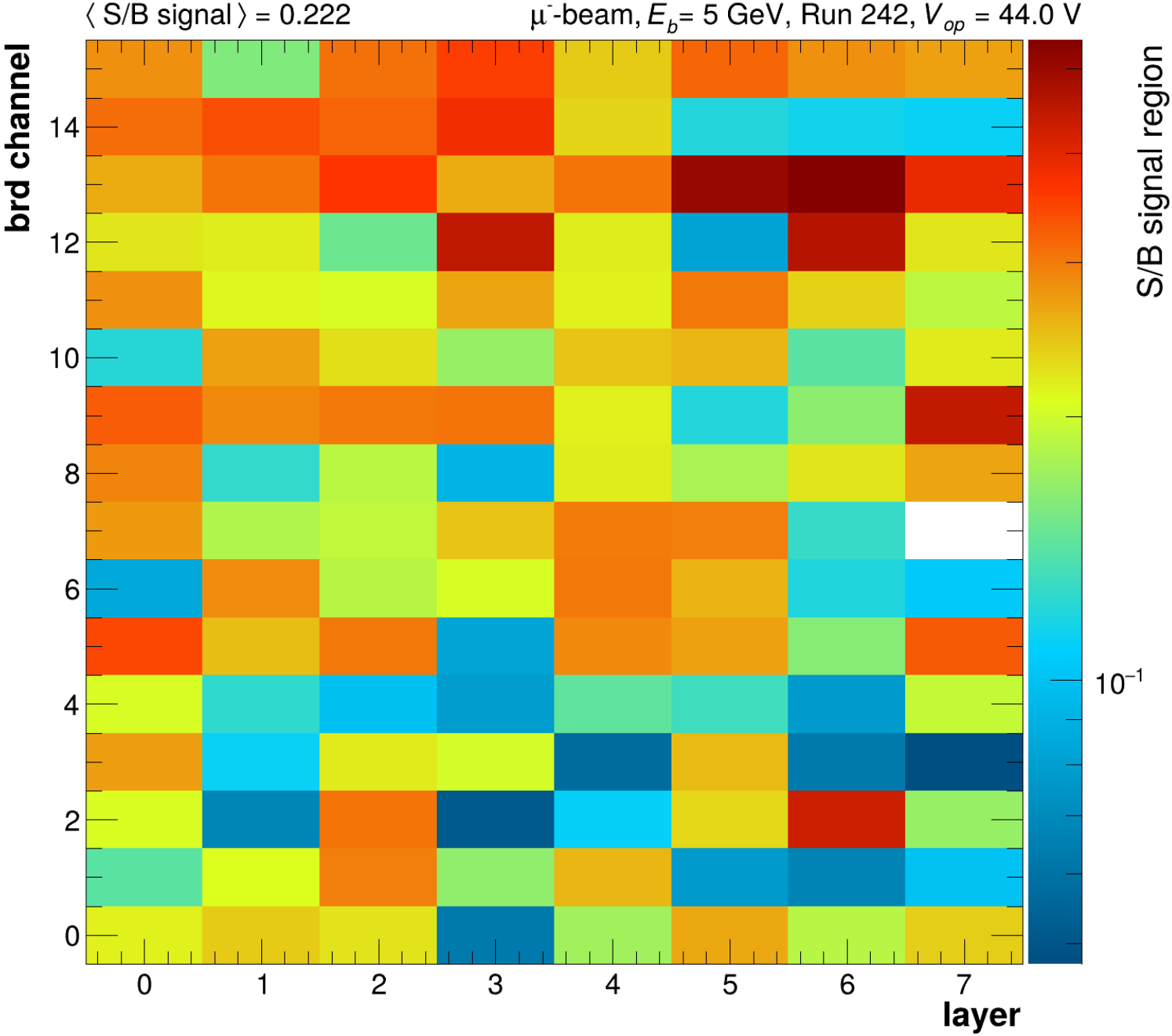

r:1 c:0 m:1
Total events: 510074
Landau-Gauss fit, trigg.
L MPV = 42.13 ± 0.18
Max = 45.40 ± 0.19
χ²/ndf = 17.81
SNR = 0.78
r:1 c:1 m:1
Total events: 501761
Landau-Gauss fit, trigg.
L MPV = 42.22 ± 0.14
Max = 45.41 ± 0.15
χ²/ndf = 28.21
SNR = 0.72
r:1 c:2 m:1
Total events: 498226
Landau-Gauss fit, trigg.
L MPV = 39.57 ± 0.14
Max = 42.61 ± 0.15
χ²/ndf = 34.16
SNR = 1.66
r:1 c:3 m:1 layer 7
Total events: 491922
Landau-Gauss fit, trigg.
L MPV = 41.52 ± 0.17
Max = 44.71 ± 0.19
χ²/ndf = 22.02
SNR = 0.75
μ⁻-beam, E_b = 5 GeV
Run 242, V_{op} = 44.0 V
r:0 c:0 m:1
Total events: 506397
Landau-Gauss fit, trigg.
L MPV = 40.40 ± 0.18
Max = 43.51 ± 0.19
χ²/ndf = 12.74
SNR = 0.60
r:0 c:1 m:1
Total events: 501459
Landau-Gauss fit, trigg.
L MPV = 42.51 ± 0.15
Max = 45.73 ± 0.16
χ²/ndf = 26.21
SNR = 0.54
r:0 c:2 m:1
Total events: 445476
Landau-Gauss fit, trigg.
L MPV = 42.05 ± 0.17
Max = 45.53 ± 0.19
χ²/ndf = 22.30
SNR = 0.77
r:0 c:3 m:1
Total events: 293851
Landau-Gauss fit, trigg.
L MPV = 44.35 ± 0.23
Max = 47.92 ± 0.25
χ²/ndf = 17.33
SNR = 0.71
r:1 c:0 m:0
Total events: 773098
Landau-Gauss fit, trigg.
L MPV = 40.00 ± 0.17
Max = 43.13 ± 0.19
χ²/ndf = 15.94
SNR = 0.75
r:1 c:1 m:0
Total events: 771454
Landau-Gauss fit, trigg.
L MPV = 42.47 ± 0.14
Max = 45.76 ± 0.15
χ²/ndf = 30.04
SNR = 0.76
r:1 c:2 m:0
Total events: 771115
Landau-Gauss fit, trigg.
L MPV = 42.09 ± 0.14
Max = 45.37 ± 0.16
χ²/ndf = 32.99
SNR = 1.17
r:1 c:3 m:0
r:0 c:0 m:0
Total events: 772633
Landau-Gauss fit, trigg.
L MPV = 40.06 ± 0.19
Max = 43.23 ± 0.20
χ²/ndf = 8.95
SNR = 0.74
r:0 c:1 m:0
Total events: 771295
Landau-Gauss fit, trigg.
L MPV = 41.17 ± 0.16
Max = 44.45 ± 0.17
χ²/ndf = 18.07
SNR = 1.07
r:0 c:2 m:0
Total events: 747611
Landau-Gauss fit, trigg.
L MPV = 42.93 ± 0.20
Max = 46.41 ± 0.22
χ²/ndf = 34.99
SNR = 0.95
r:0 c:3 m:0
Total events: 662697
Landau-Gauss fit, trigg.
L MPV = 43.13 ± 0.24
Max = 46.64 ± 0.25
χ²/ndf = 14.40
SNR = 1.92
counts
ADC (arb. units)

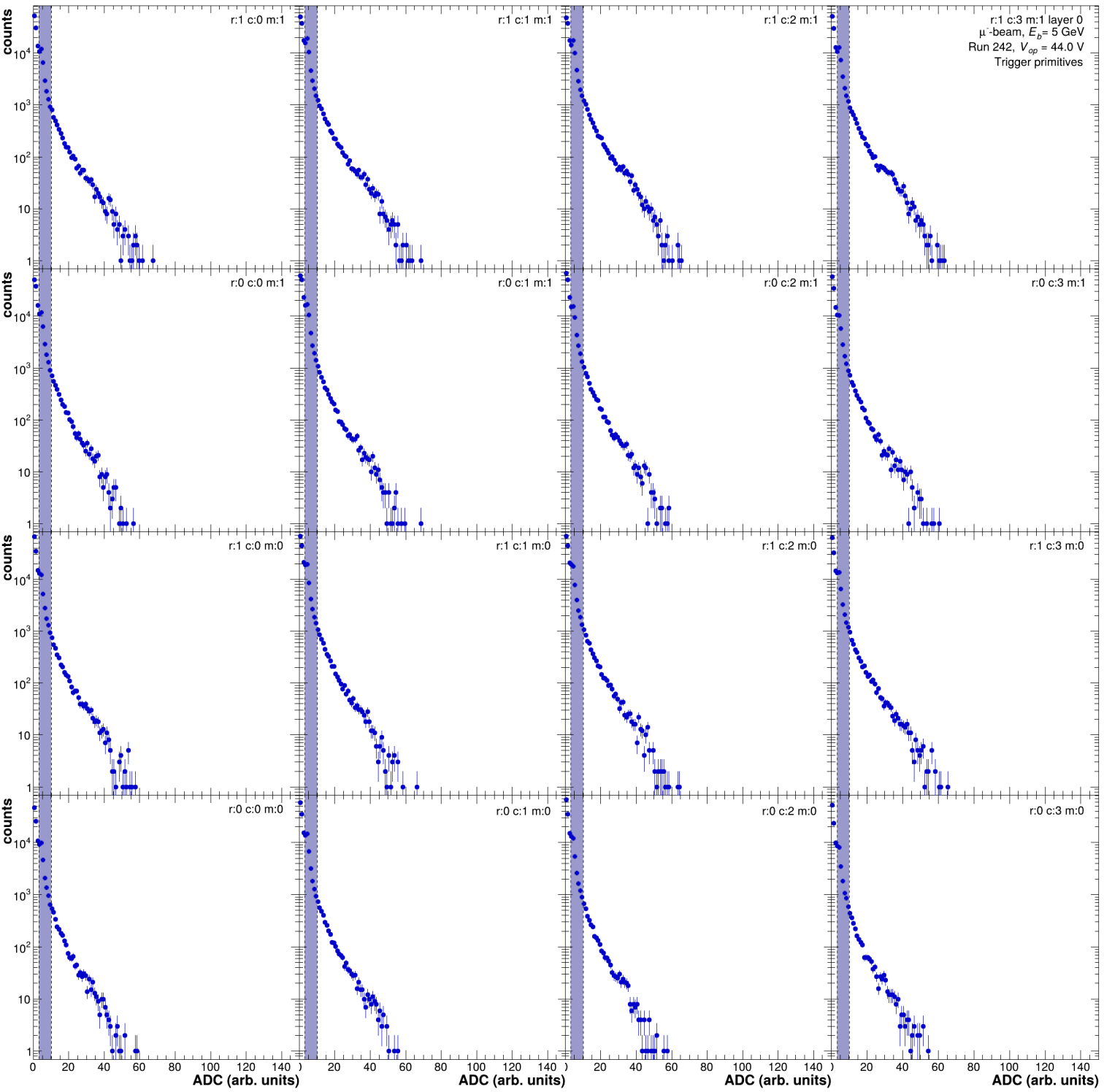

μ⁻-beam, E_b= 5 GeV, Run 242, V_{op} = 44.0 V
Y (row)
X (col)
Num triggers
×10³
3.5
3.0
2.5
2.0
1.5
1.0
0.5
0.0
−0.5
−0.5
0
0.5
1
1.5
2
2.5
3
3.5
400
350
300
250
200

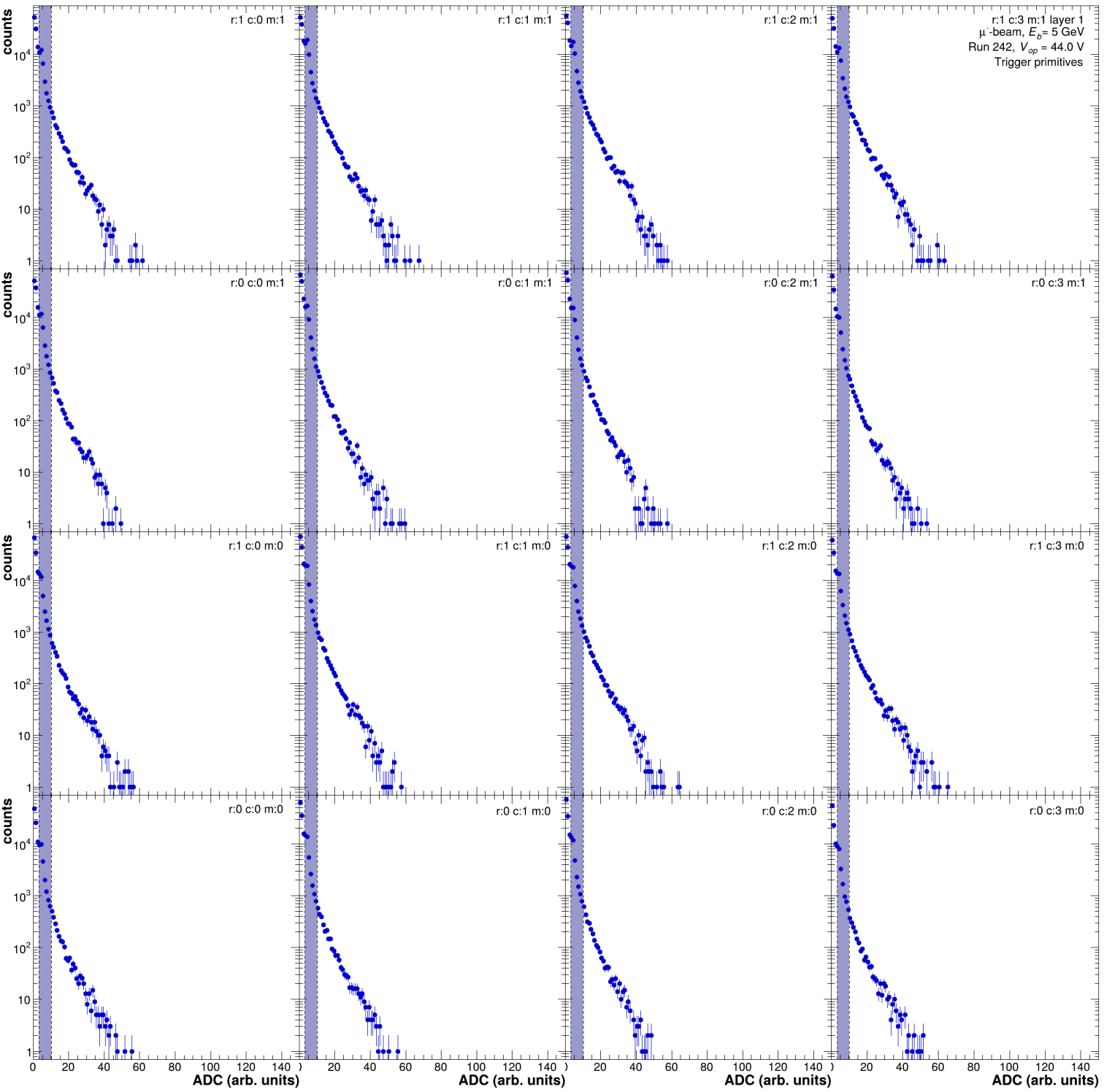

evt. collected = 773211
μ⁻-beam, E_b= 5 GeV, Run 242, V_{op} = 44.0 V
brd channel
0
2
4
6
8
10
12
14
0
1
2
3
4
5
6
7
layer
counts

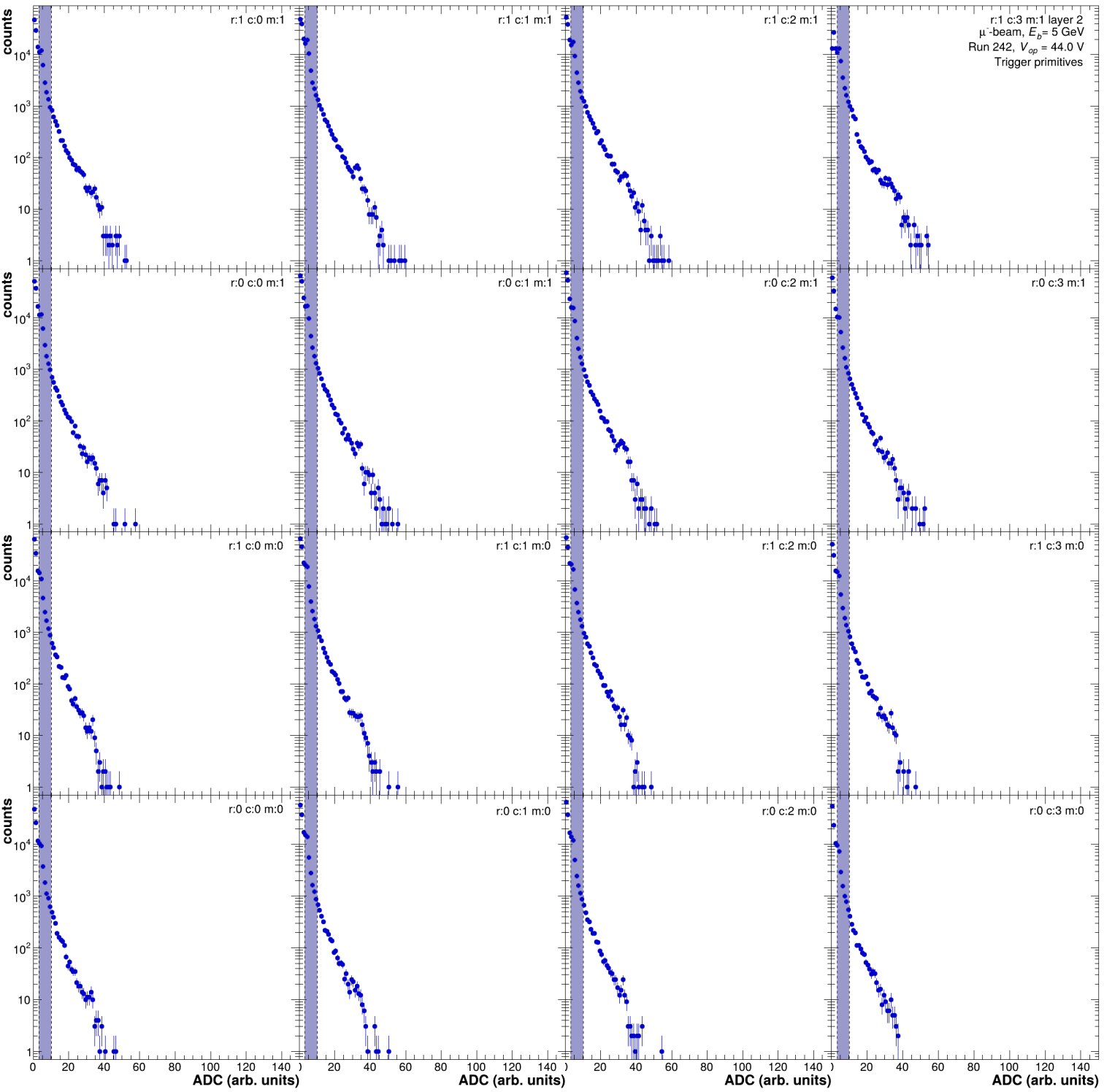

μ⁻-beam, E_b= 5 GeV, Run 242, V_{op} = 44.0 V
brd channel
0
2
4
6
8
10
12
14
layers
0
1
2
3
4
5
6
7
SNR
2
4
6
8
10

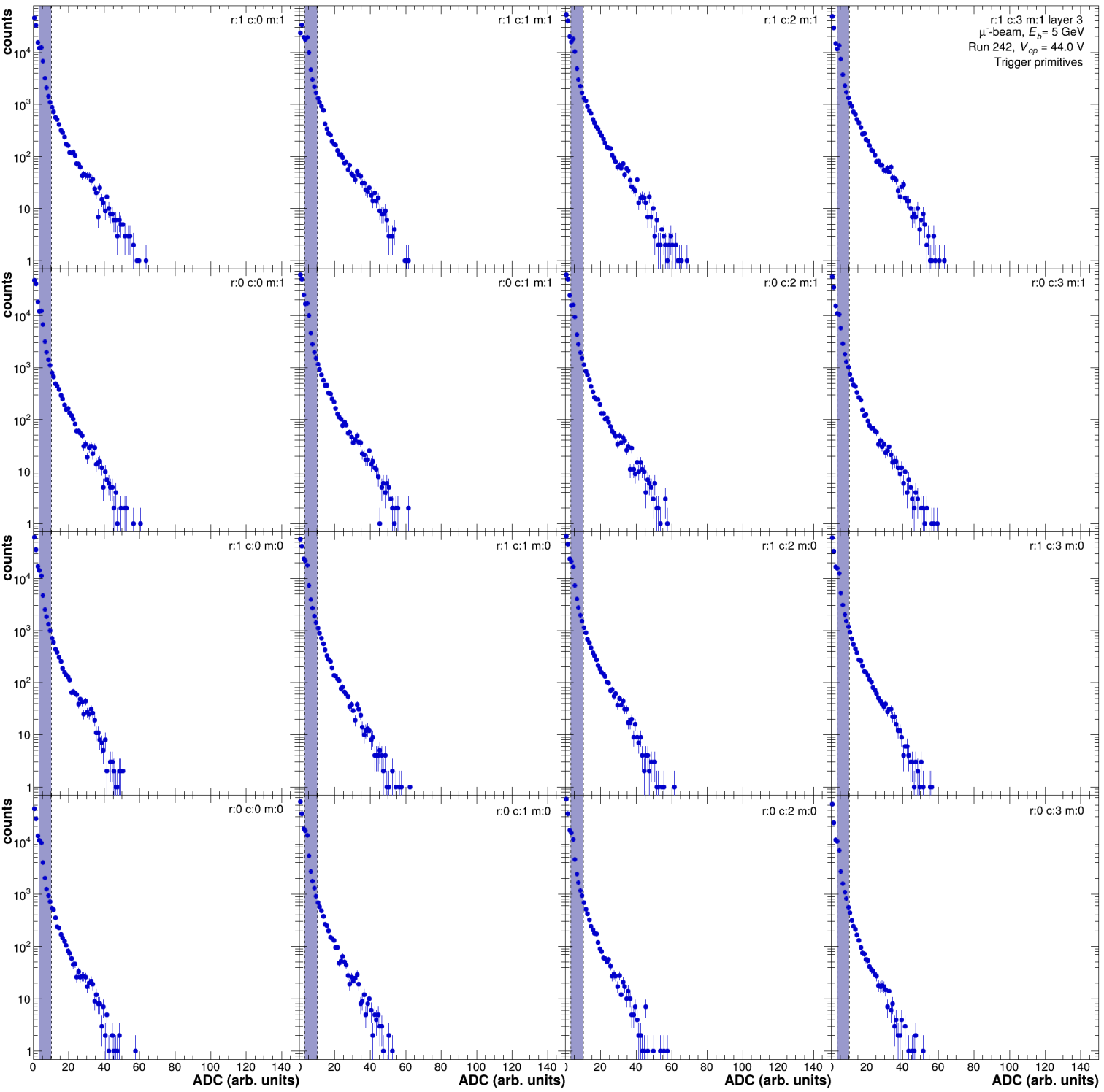

⟨ S/B noise ⟩ = 0.029
μ⁻-beam, E_b= 5 GeV, Run 242, V_{op} = 44.0 V
brd channel
0
2
4
6
8
10
12
14
0
1
2
3
4
5
6
7
layer
S/B noise region
10^{-1}
10^{-2}

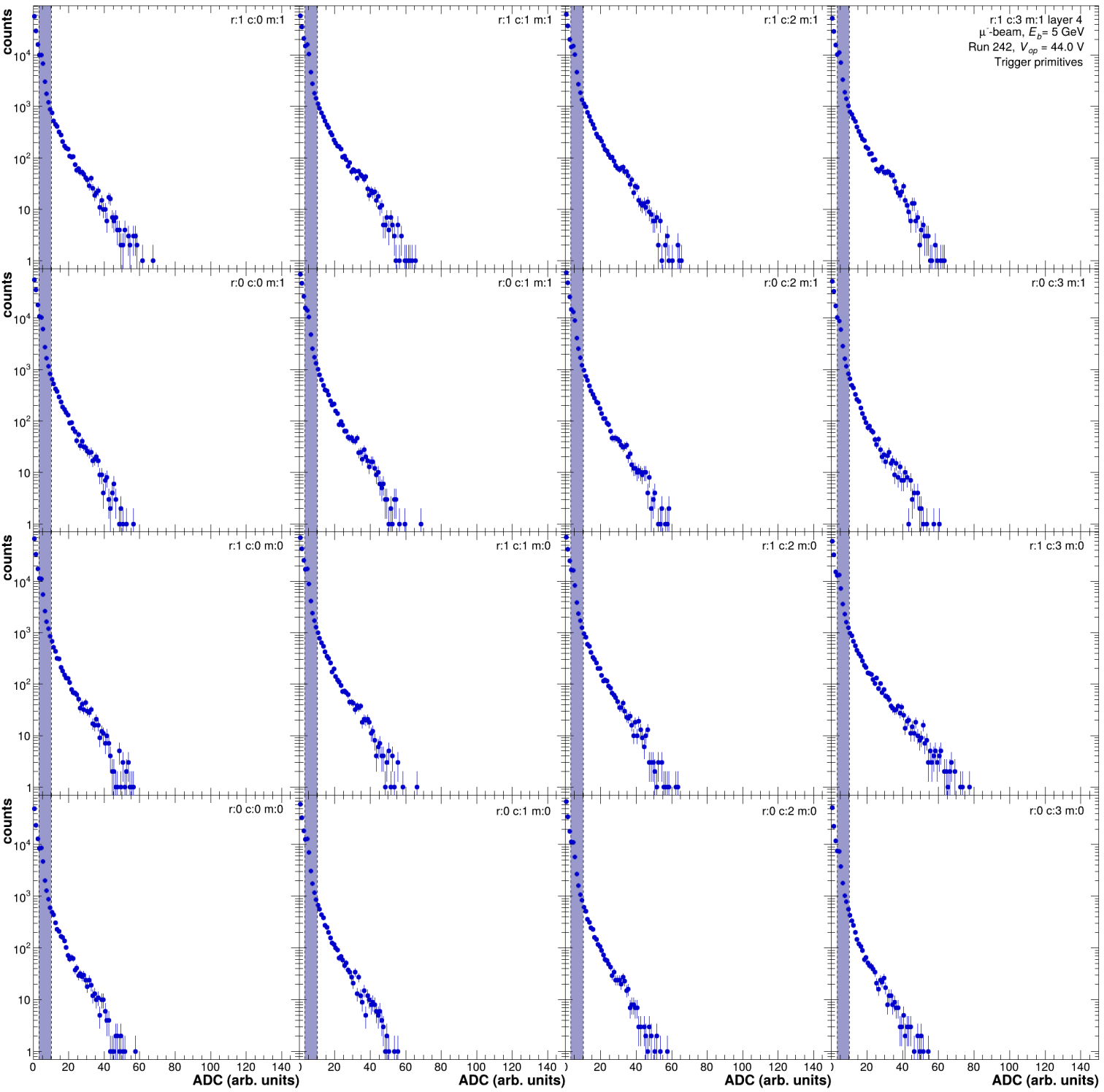

⟨ S/B signal ⟩ = 0.222
μ⁻-beam, E_b= 5 GeV, Run 242, V_{op} = 44.0 V
brd channel
0
2
4
6
8
10
12
14
0
1
2
3
4
5
6
7
layer
S/B signal region
10^{-1}

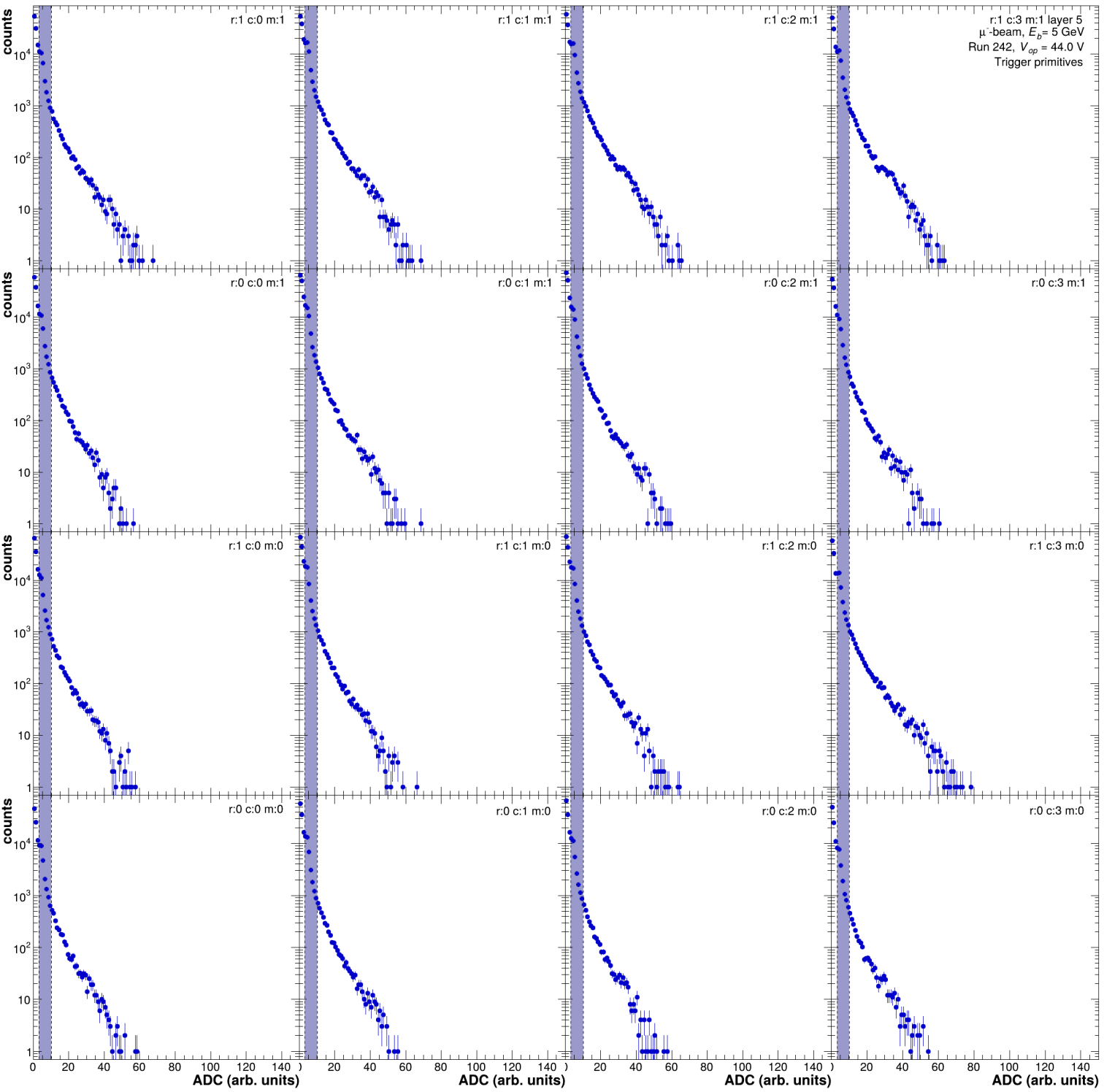

r:1 c:0 m:1
r:1 c:1 m:1
r:1 c:2 m:1
r:1 c:3 m:1 layer 0
μ⁻-beam, E_b= 5 GeV
Run 242, V_{op} = 44.0 V
Trigger primitives
r:0 c:0 m:1
r:0 c:1 m:1
r:0 c:2 m:1
r:0 c:3 m:1
r:1 c:0 m:0
r:1 c:1 m:0
r:1 c:2 m:0
r:1 c:3 m:0
r:0 c:0 m:0
r:0 c:1 m:0
r:0 c:2 m:0
r:0 c:3 m:0
counts
ADC (arb. units)

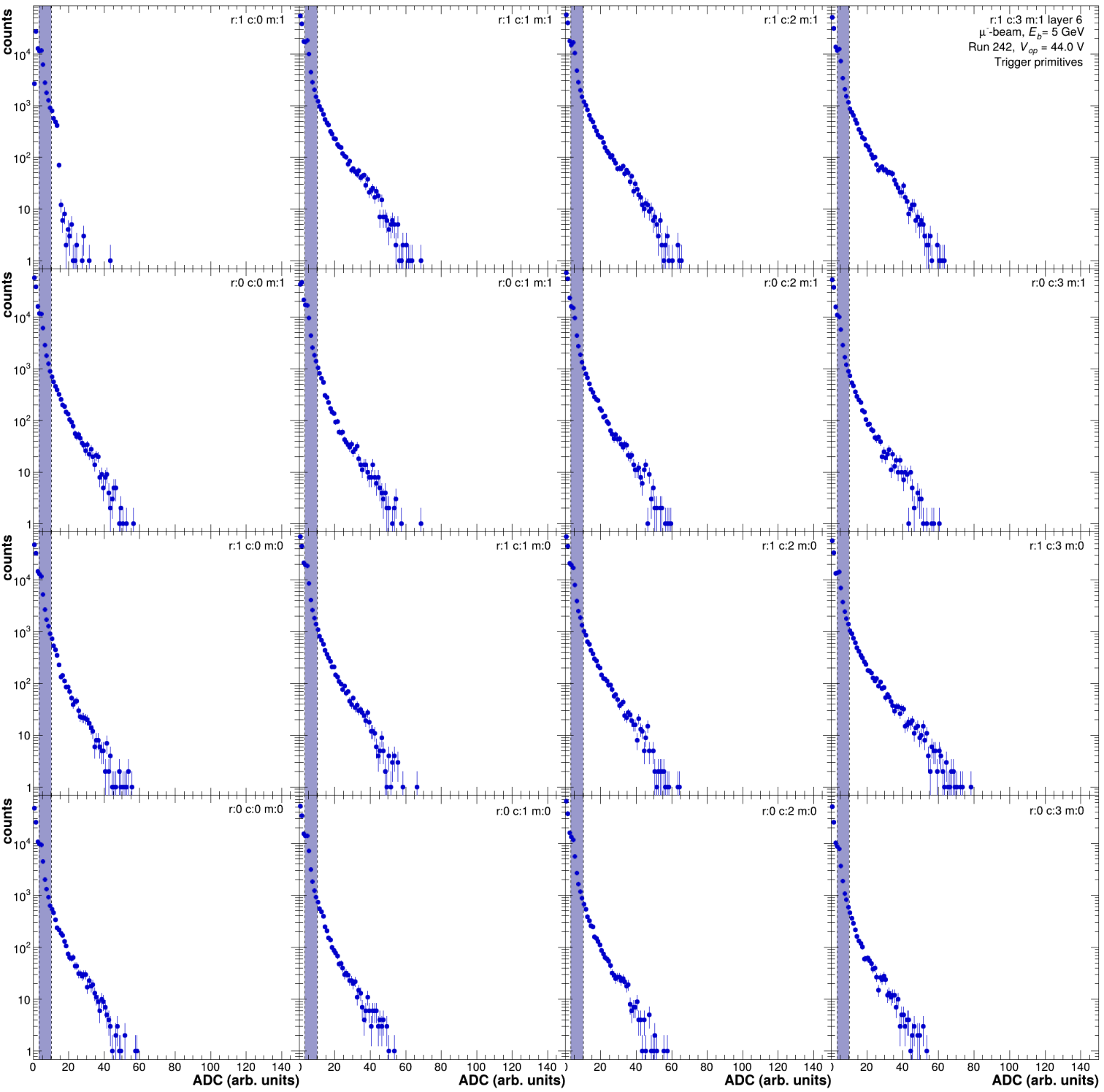

r:1 c:0 m:1
r:1 c:1 m:1
r:1 c:2 m:1
r:1 c:3 m:1 layer 1
μ⁻-beam, E_b= 5 GeV
Run 242, V_{op} = 44.0 V
Trigger primitives
r:0 c:0 m:1
r:0 c:1 m:1
r:0 c:2 m:1
r:0 c:3 m:1
r:1 c:0 m:0
r:1 c:1 m:0
r:1 c:2 m:0
r:1 c:3 m:0
r:0 c:0 m:0
r:0 c:1 m:0
r:0 c:2 m:0
r:0 c:3 m:0
counts
ADC (arb. units)

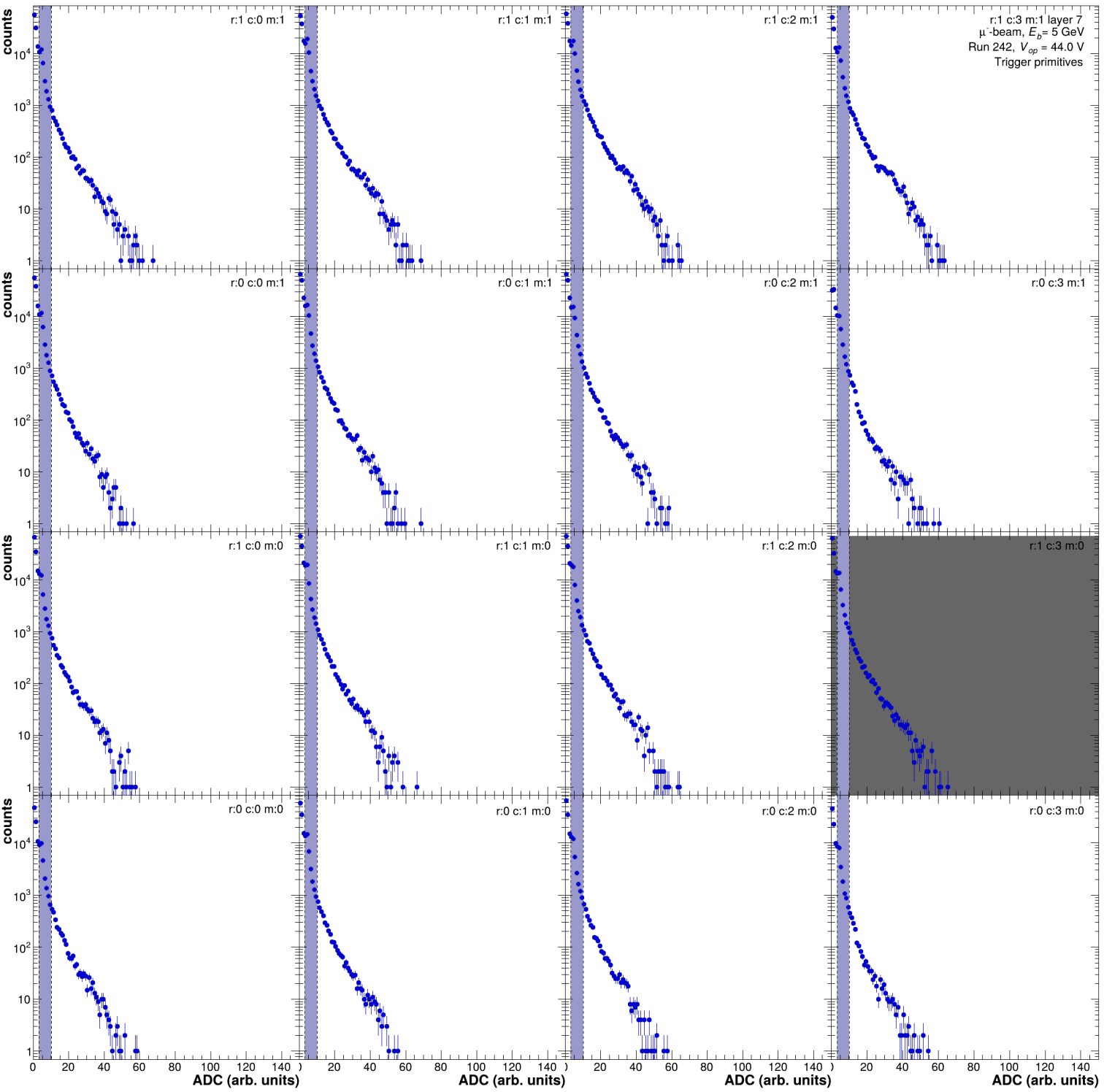

r:1 c:0 m:1
r:1 c:1 m:1
r:1 c:2 m:1
r:1 c:3 m:1 layer 2
μ⁻-beam, E_b= 5 GeV
Run 242, V_{op} = 44.0 V
Trigger primitives
r:0 c:0 m:1
r:0 c:1 m:1
r:0 c:2 m:1
r:0 c:3 m:1
r:1 c:0 m:0
r:1 c:1 m:0
r:1 c:2 m:0
r:1 c:3 m:0
r:0 c:0 m:0
r:0 c:1 m:0
r:0 c:2 m:0
r:0 c:3 m:0
counts
ADC (arb. units)


r:1 c:0 m:1
r:1 c:1 m:1
r:1 c:2 m:1
r:1 c:3 m:1 layer 3
μ⁻-beam, E_b= 5 GeV
Run 242, V_{op} = 44.0 V
Trigger primitives
r:0 c:0 m:1
r:0 c:1 m:1
r:0 c:2 m:1
r:0 c:3 m:1
r:1 c:0 m:0
r:1 c:1 m:0
r:1 c:2 m:0
r:1 c:3 m:0
r:0 c:0 m:0
r:0 c:1 m:0
r:0 c:2 m:0
r:0 c:3 m:0
counts
ADC (arb. units)


r:1 c:0 m:1
r:1 c:1 m:1
r:1 c:2 m:1
r:1 c:3 m:1 layer 4
μ⁻-beam, E_b= 5 GeV
Run 242, V_{op} = 44.0 V
Trigger primitives
r:0 c:0 m:1
r:0 c:1 m:1
r:0 c:2 m:1
r:0 c:3 m:1
r:1 c:0 m:0
r:1 c:1 m:0
r:1 c:2 m:0
r:1 c:3 m:0
r:0 c:0 m:0
r:0 c:1 m:0
r:0 c:2 m:0
r:0 c:3 m:0
counts
ADC (arb. units)


r:1 c:0 m:1
r:1 c:1 m:1
r:1 c:2 m:1
r:1 c:3 m:1 layer 5
μ⁻-beam, E_b= 5 GeV
Run 242, V_{op} = 44.0 V
Trigger primitives
r:0 c:0 m:1
r:0 c:1 m:1
r:0 c:2 m:1
r:0 c:3 m:1
r:1 c:0 m:0
r:1 c:1 m:0
r:1 c:2 m:0
r:1 c:3 m:0
r:0 c:0 m:0
r:0 c:1 m:0
r:0 c:2 m:0
r:0 c:3 m:0
counts
ADC (arb. units)


r:1 c:0 m:1
r:1 c:1 m:1
r:1 c:2 m:1
r:1 c:3 m:1 layer 6
μ⁻-beam, E_b= 5 GeV
Run 242, V_{op} = 44.0 V
Trigger primitives
r:0 c:0 m:1
r:0 c:1 m:1
r:0 c:2 m:1
r:0 c:3 m:1
r:1 c:0 m:0
r:1 c:1 m:0
r:1 c:2 m:0
r:1 c:3 m:0
r:0 c:0 m:0
r:0 c:1 m:0
r:0 c:2 m:0
r:0 c:3 m:0
counts
ADC (arb. units)


r:1 c:0 m:1
r:1 c:1 m:1
r:1 c:2 m:1
r:1 c:3 m:1 layer 7
μ⁻-beam, E_b= 5 GeV
Run 242, V_{op} = 44.0 V
Trigger primitives
r:0 c:0 m:1
r:0 c:1 m:1
r:0 c:2 m:1
r:0 c:3 m:1
r:1 c:0 m:0
r:1 c:1 m:0
r:1 c:2 m:0
r:1 c:3 m:0
r:0 c:0 m:0
r:0 c:1 m:0
r:0 c:2 m:0
r:0 c:3 m:0
counts
ADC (arb. units)